

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407872
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Sato; Kazushi

Image processing apparatus and method

Abstract

The present technology relates to an image processing apparatus and method that are capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing. The image processing apparatus includes an encoding mode setter that sets, in units of coding units having a hierarchical structure, whether a non-compression mode is to be selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data, and an encoder that encodes the image data in units of the coding units in accordance with a mode set by the encoding mode setter. The present disclosure can be applied to, for example, an image processing apparatus.

Inventors:	Sato; Kazushi (Kanagawa, JP)
Applicant:	SONY GROUP CORPORATION (Tokyo, JP)
Family ID:	46507101
Assignee:	SONY GROUP CORPORATION (Tokyo, JP)
Appl. No.:	18/738076
Filed:	June 10, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240333980 A1	Oct. 03, 2024

Foreign Application Priority Data

JP	2011-003245	Jan. 11, 2011
----	-------------	---------------

Related U.S. Application Data

continuation parent-doc US 18227591 20230728 US 12047611 child-doc US 18738076
continuation parent-doc US 17228642 20210412 ABANDONED child-doc US 18227591
continuation parent-doc US 16581575 20190924 US 10979739 20210413 child-doc US 17228642
continuation parent-doc US 15880045 20180125 US 10432976 20191001 child-doc US 16581575
continuation parent-doc US 14988365 20160105 US 10027987 20180717 child-doc US 15880045
continuation parent-doc US 13978492 US 9270987 20160223 WO PCT/JP2012/050016 20120104
child-doc US 14988365

Publication Classification

Int. Cl.: **H04N19/176** (20140101); **G06T9/00** (20060101); **H04N19/117** (20140101); **H04N19/12** (20140101); **H04N19/124** (20140101); **H04N19/132** (20140101); **H04N19/152** (20140101); **H04N19/157** (20140101); **H04N19/184** (20140101); **H04N19/30** (20140101); **H04N19/44** (20140101); **H04N19/46** (20140101); **H04N19/70** (20140101); **H04N19/82** (20140101); **H04N19/85** (20140101)

U.S. Cl.:

CPC **H04N19/82** (20141101); **G06T9/007** (20130101); **H04N19/117** (20141101); **H04N19/12** (20141101); **H04N19/124** (20141101); **H04N19/132** (20141101); **H04N19/152** (20141101); **H04N19/157** (20141101); **H04N19/176** (20141101); **H04N19/184** (20141101); **H04N19/30** (20141101); **H04N19/44** (20141101); **H04N19/46** (20141101); **H04N19/70** (20141101); **H04N19/85** (20141101);

Field of Classification Search

CPC: G06T (9/007); H04N (19/117); H04N (19/12); H04N (19/124); H04N (19/132); H04N (19/152); H04N (19/157); H04N (19/176); H04N (19/184); H04N (19/30); H04N (19/44); H04N (19/46); H04N (19/70); H04N (19/82); H04N (19/85)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5335116	12/1993	Onishi et al.	N/A	N/A
5485283	12/1995	Kaneko	N/A	N/A
5485293	12/1995	Robinder	N/A	N/A
5661573	12/1996	Takehara et al.	N/A	N/A
5661576	12/1996	Kaneko	N/A	N/A
6671413	12/2002	Pearlman et al.	N/A	N/A
6965700	12/2004	Pearlman et al.	N/A	N/A
7936931	12/2010	Mizuno	N/A	N/A
8503809	12/2012	Fukuhara et al.	N/A	N/A
8737751	12/2013	Sato	N/A	N/A
8768077	12/2013	Sato	N/A	N/A
9270987	12/2015	Sato	N/A	N/A
9838716	12/2016	Sato	N/A	N/A

10027987	12/2017	Sato	N/A	H04N 19/82
12047611	12/2023	Sato	N/A	H04N 19/152
2001/0043731	12/2000	Ito et al.	N/A	N/A
2004/0057514	12/2003	Kishi	N/A	N/A
2004/0175048	12/2003	Pearlman et al.	N/A	N/A
2005/0249289	12/2004	Yagasaki et al.	N/A	N/A
2007/0133892	12/2006	Chiba et al.	N/A	N/A
2007/0223585	12/2006	Fujisawa et al.	N/A	N/A
2007/0230572	12/2006	Koto et al.	N/A	N/A
2007/0237412	12/2006	Mizuno	N/A	N/A
2007/0280353	12/2006	Arakawa et al.	N/A	N/A
2009/0016626	12/2008	Zhang et al.	N/A	N/A
2009/0103530	12/2008	Fartmann et al.	N/A	N/A
2009/0154560	12/2008	Hong et al.	N/A	N/A
2009/0175548	12/2008	Fukuhara et al.	N/A	N/A
2009/0274209	12/2008	Garg et al.	N/A	N/A
2009/0316788	12/2008	Techernatinsky et al.	N/A	N/A
2010/0086029	12/2009	Chen et al.	N/A	N/A
2010/0091840	12/2009	Gao et al.	N/A	N/A
2012/0287993	12/2011	Priddle et al.	N/A	N/A
2013/0003871	12/2012	Bjontegaard et al.	N/A	N/A
2013/0077682	12/2012	Han et al.	N/A	N/A
2013/0089265	12/2012	Yie et al.	N/A	N/A
2013/0223518	12/2012	Shibahara et al.	N/A	N/A
2013/0329783	12/2012	Sato	N/A	N/A
2014/0140405	12/2013	Seregin et al.	N/A	N/A
2014/0355679	12/2013	Rosewarne	N/A	N/A
2015/0103901	12/2014	Sato	N/A	N/A
2016/0134897	12/2015	Sato	N/A	N/A
2016/0134898	12/2015	Sato	N/A	N/A
2017/0034525	12/2016	Sato	N/A	N/A
2021/0306674	12/2020	Sato	N/A	H04N 19/46
2023/0388554	12/2022	Sato	N/A	H04N 19/132

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101031086	12/2006	CN	N/A
101577110	12/2008	CN	N/A
3308921	12/2001	JP	N/A
2004-135251	12/2003	JP	N/A
2007-166039	12/2006	JP	N/A
3992303	12/2006	JP	N/A
4240283	12/2008	JP	N/A
2010-193398	12/2009	JP	N/A

10-2010-0111609	12/2019	KR	N/A
WO 2004/034330	12/2003	WO	N/A
WO 2009/011279	12/2008	WO	N/A
WO 2009/031648	12/2008	WO	N/A
WO 2010/002214	12/2009	WO	N/A
WO 2012/008130	12/2011	WO	N/A
WO 2012/070232	12/2011	WO	N/A

OTHER PUBLICATIONS

Oct. 8, 2015, Chinese Office Action for related CN application No. 201280004667.8. cited by applicant

Oct. 28, 2015, Singaporean Office Action for related SG application No. 2014014526. cited by applicant

Jul. 23, 2015, JP communication issued for related JP application No. 2011-003245. cited by applicant

Thomas Wiegand, Editor's Proposed Draft Text Modifications for Joint Video Specification (ITU-T Rec. H.264 ISO/IEC 14496-10 AVC), Geneva modifications draft 37, Joint Video Team (JVT) of ISO/IEC MPEG and ITU-T VCEG, Oct. 9-17, 2002, pp. i-122, 5.SUP.th .Meeting: Geneva, CH. cited by applicant

Ken McCann, et al., Samsung's Response to the Call for Proposals on Video Compression Technology, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, Apr. 15-23, 2010, 1st Meeting: Dresden, DE. cited by applicant

Jul. 23, 2015, JP communication issued for related JP application No. 2014-002802. cited by applicant

Jun. 25, 2015, EP communication issued for related EP application No. 12733885.3. cited by applicant

Apr. 16, 2015, EP communication issued for related EP application No. 12733885.3. cited by applicant

Anthony Joch, et al., UB Video comments on the draft text (JVT-E146d37), Joint Video Team (JVT) of ISO/IEC MPEG & ITU-T VCEG (ISO/IEC JTC1/SC29/WG11 and ITU-T SG16 Q.6), Dec. 5-13, 2002, pp. 1-5, 6.SUP.th .Meeting: Awaji, Island, JP. cited by applicant

Takeshi Chujoh, et al., Internal bit depth increase for coding efficiency, ITU—Telecommunications Standardization Sector, Study Group 16 Question 6, Video Coding Experts Group (VCEG), Jan. 15-16, 2007, pp. 1-6, 31.SUP.st .Meeting: Marrakech, MA. cited by applicant

H. Hu, et al., Classification-based Hybrid Filters for Image Processing, Eindhoven University of Technology, Philips Research Laboratories Eindhoven, 2006. cited by applicant

Feb. 12, 2015, JP communication issued for related JP application No. 2014-002802. cited by applicant

Feb. 12, 2015, JP communication issued for related JP application No. 2011-003245. cited by applicant

Mar. 9, 2015, SG communication issued for related SG application No. 2013052105. cited by applicant

Sep. 30, 2014, JP communication issued for related JP application No. 2014-002802. cited by applicant

Sep. 30, 2014, JP communication issued for related JP application No. 2011-003245. cited by applicant

Takeshi Chujoh, et al., "Block-based Adaptive Loop Filter", Jul. 16-18, 2008, pp. 1-6, ITU—Telecommunications Sector, Video-Coding Experts Group (VCEG), 35.SUP.th .Meeting: Berlin, Germany. cited by applicant

"Test Model under Consideration", Jul. 21-28, 2010, pp. 1-198, Joint Collaborative Team on Video

Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, 2.SUP.nd .Meeting: Geneva, Switzerland. cited by applicant

Takeshi Chujoh, et al., “Internal bit depth increase except frame memory”, Apr. 20-21, 2007, pp. 1-4, ITU—Telecommunications Standardization Sector, Video Coding Experts Group (VCEG), 32.SUP.nd .Meeting: San Jose, USA. cited by applicant

Keiichi Chono, et al., “Pulse code modulation mode for HEVC”, Jan. 20-28, 2011, pp. 1-9, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, 4.SUP.th .Meeting: Daegu, Korea. cited by applicant

Gary Sullivan, “Seven Steps Toward a More Robust Codec Design”, May 6-10, 2002, pp. 1-7, Joint Video Team (JVT) of ISO/IEC MPEG & ITU-T VCEG (ISO/IEC JTC1/SC29/WG11 and ITU-T SG16 Q.6), 3.SUP.rd .Meeting: Fairfax, Virginia, USA. cited by applicant

Keiichi Chono, et al., “Proposal of enhanced PCM coding in HEVC”, Mar. 16-23, 2011, pp. 1-12, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, 5.SUP.th .Meeting: Geneva, Switzerland. cited by applicant

Dec. 20, 2016, JP communication issued for related JP application No. 2015-229616. cited by applicant

Dec. 20, 2016, JP communication issued for related JP application No. 2015-229617. cited by applicant

Nov. 29, 2017, Korean Office Action issued for related KR application No. 10-2013-7017198. cited by applicant

Sprljan et al., TE3 subtest 3: Local intensity compensation (LIC) for inter prediction, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG11, Oct. 7-15, 2010, pp. 1-4, 3rd Meeting: Guangzhou, CN. cited by applicant

Jun. 6, 2018, European Communication issued for related EP application No. 12733885.3. cited by applicant

Apr. 10, 2018, Singaporean Office Action issued for related SG Application No. 10201605708X. cited by applicant

Apr. 13, 2018, Korean Office Action issued for related KR Application No. 10-2018-7002833. cited by applicant

Apr. 13, 2018, Korean Office Action issued for related KR Application No. 10-2018-7002831. cited by applicant

Mar. 21, 2018, Korean Office Action issued for related KR application No. 10-2018-7002830. cited by applicant

Feb. 1, 2018, Japanese Office Action issued for related JP application No. 2017-040115. cited by applicant

Dec. 12, 2017, Japanese Office Action issued for related JP application No. 2017-040116. cited by applicant

Chujoh et al., Specification and Experimental Results of Quadtree-Based Adaptive Loop Filter, ITU—Telecommunications Standardization Sector Study Group 16 Question 6 Video Coding Experts Group (VCEG) 3th Meeting, Apr. 15-18, 2009, pp. 1-10, Yokohama, Japan. cited by applicant

Dec. 12, 2017, Japanese Office Action Issued for related JP application No. 2017-040115. cited by applicant

First Office Action received in Chinese Application No. 201610839366.7, mailed Jan. 22, 2019 12 pages. cited by applicant

Feb. 8, 2018, Singaporean Written Opinion issued for related SG application No. 10201605708X. cited by applicant

Background/Summary

CROSS REFERENCE TO PRIOR APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 18/227,591 (filed on Jul. 28, 2023), which is a continuation of U.S. patent application Ser. No. 17/228,642 (filed on Apr. 12, 2021), which is a continuation of U.S. patent application Ser. No. 16/581,575 (filed on Sep. 24, 2019 and issued as U.S. Pat. No. 10,979,739 on Apr. 13, 2021), which is a continuation of U.S. patent application Ser. No. 15/880,045 (filed on Jan. 25, 2018 and issued as U.S. patent No. 10,432,976 on Oct. 1, 2019), which is a continuation of U.S. patent application Ser. No. 14/988,365 (filed on Jan. 5, 2016 and issued as U.S. Pat. No. 10,027,987 on Jul. 17, 2018), which is a continuation of U.S. patent application Ser. No. 13/978,492 (filed on Aug. 6, 2013 and issued as U.S. Pat. No. 9,270,987 on Feb. 23, 2016) which is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2012/050016 (filed on Jan. 4, 2012) under 35 U.S.C. § 371, which claims priority to Japanese patent application Ser. No. 2011-003245 (filed on Jan. 11, 2011), which are all hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to an image processing apparatus and method, and particularly relates to an image processing apparatus and method that enable enhancement of encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

BACKGROUND ART

(2) In recent years, apparatuses compliant with a compression format such as MPEG (Moving Picture Experts Group), in which image information is digitally handled and is compressed using an orthogonal transform such as the discrete cosine transform and by motion compensation by utilizing a redundancy particular to image information for the purpose of achieving highly efficient transmission and accumulation of information, have been becoming prevalent for use in both distribution of information from broadcast stations and reception of information in general households.

(3) In particular, MPEG-2 (ISO (International Organization for Standardization)/IEC (International Electrotechnical Commission) 13818-2) is defined as a general-purpose image coding format, is a standard that covers both interlaced scanned images and progressive scanned images as well as standard-definition images and high-definition images, and is currently widely used in a broad range of applications for professional use and consumer use. With the MPEG-2 compression format, a high compression rate and a favorable image quality can be realized by, for example, allocating an amount of code (bit rate) of 4 to 8 Mbps to a standard-definition interlaced scanned image having 720×480 pixels or an amount of code (bit rate) of 18 to 22 Mbps to a high-definition interlaced scanned image having 1920×1088 pixels.

(4) MPEG-2 is mainly used for high image quality encoding suitable for broadcasting, but is not compatible with coding formats of an amount of code (bit rate) lower than that of MPEG-1, that is, a higher compression rate. With the widespread use of mobile terminals, the need for such coding formats will increase in the future, and in response the MPEG-4 coding format has been standardized. Regarding an image coding format, MPEG-4 was designated an international standard as ISO/IEC 14496-2 in December 1998.

(5) Furthermore, standardization of the format H.26L (ITU-T (International Telecommunication Union Telecommunication Standardization Sector) Q6/16 VCEG (Video Coding Expert Group)), which was initially for the purpose of image coding for video conferences, has been progressing in recent years. It is known that H.26L realizes higher encoding efficiency than previous coding

formats, such as MPEG-2 and MPEG-4, though encoding and decoding according to H.26L involve a larger amount of computation. Also, as part of MPEG-4 activities, standardization for realizing higher encoding efficiency by introducing functions which are not supported by H.26L on the basis of H.26L is currently progressing as Joint Model of Enhanced-Compression Video Coding.

(6) As the standardization schedule, a standard called H.264 and MPEG-4 Part 10 (Advanced Video Coding, hereinafter referred to as AVC) was designated an international standard in March 2003.

(7) Furthermore, as an extension of the above, standardization of FReXt (Fidelity Range Extension), including encoding tools necessary for business use, such as RGB, 4:2:2, and 4:4:4, as well as 8×8 DCT (Discrete Cosine Transform) and quantization matrices defined in MPEG-2, was completed in February 2005. Accordingly, a coding format capable of favorably expressing even film noise included in movies by using AVC has been established, which is used for a wide range of applications such as Blu-Ray Discs.

(8) However, there has recently been a growing need for encoding at a higher compression rate, for example, compression of an image having about 4000×2000 pixels, which is four times the number of pixels included in a high-definition image, or distribution of high-definition images in an environment with a limited transmission capacity, such as the Internet. Therefore, in VCEG (Video Coding Expert Group) under ITU-T, ongoing studies for enhancing encoding efficiency have been performed.

(9) Meanwhile, for the purpose of realizing higher encoding efficiency than that of AVC, standardization of a coding format called HEVC (High Efficiency Video Coding) is currently being conducted by JCTVC (Joint Collaboration Team-Video Coding), which is a standards group of ITU-T and ISO/IEC (see, for example, NPL 1).

(10) In the HEVC coding format, coding units (CUs) are defined as units of processing which are similar to macroblocks used in AVC. Unlike a macroblock used in AVC, the size of a CU is not fixed to 16×16 pixels, and is specified in image compression information in each sequence.

(11) CUs are hierarchically configured from a largest coding unit (LCU) to a smallest coding unit (SCU). That is, it may be generally considered that an LCU corresponds to a macroblock used in AVC, and a CU in a layer lower than the LCU corresponds to a sub-macroblock used in AVC.

(12) Meanwhile, there is a coding format in which an encoding mode for encoding and outputting image data and a non-encoding mode for outputting image data without encoding the image data are provided, whether the encoding mode or the non-encoding mode is to be used is selected in units of macroblocks, and the encoding mode and the non-encoding mode can be used in combination within a single picture (see, for example, PTL 1). Also in the AVC coding format, an I_PCM mode for outputting image data without encoding the image data is supported as mb_type (see, for example, PTL 2). This is used for ensuring real-time operation of arithmetic coding processing in a case where a quantization parameter is set to be a small value, such as QP=0, and in a case where the information amount of encoded data is larger than that of an input image. Also, it is possible to realize lossless coding by using I-PCM.

(13) Also, a method for increasing internal arithmetic has been suggested (see, for example, NPL 2). Accordingly, an internal arithmetic error caused in processing such as an orthogonal transform and motion compensation can be reduced, and encoding efficiency can be enhanced.

(14) Furthermore, a technique in which an FIR filter is provided in a motion compensation loop has been suggested (see, for example, NPL 3). In an encoding apparatus, by obtaining an FIR filter coefficient using a Wiener filter so as to minimize an error with respect to an input image, degradation in a reference image can be minimized, and encoding efficiency of image compression information to be output can be enhanced.

CITATION LIST

Patent Literature

(15) PTL 1: Japanese Patent No. 3992303 PTL 2: Japanese Patent No. 4240283

(16) NPL 1: "Test Model under Consideration", JCTVC-B205, Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG16 WP3 and ISO/IEC JTC1/SC29/WG112nd Meeting: Geneva, CH, 21-28 Jul. 2010 NPL 2: Takeshi Chujoh, Reiko Noda, "Internal bit depth increase except frame memory", VCEG-AF07, ITU—Telecommunications Standardization Sector STUDY GROUP 16 Question 6 Video Coding Experts Group (VCEG) 32nd Meeting: San Jose, USA, 20-21 Apr. 2007 NPL 3: Takeshi Chujoh, Goki Yasuda, Naofumi Wada, Takashi Watanabe, Tomoo Yamakage, "Block-based Adaptive Loop Filter", VCEG-AI18, ITU—Telecommunications Standardization Sector STUDY GROUP 16 Question 6 Video Coding Experts Group (VCEG) 35th Meeting: Berlin, Germany, 16-18 Jul. 2008

SUMMARY OF INVENTION

Technical Problem

(17) However, in the case of a coding format in which CUs are defined as in HEVC and various processing operations are performed in units of CUs, it is considered that a macroblock used in AVC corresponds to an LCU, but if I_PCM can be set only in units of LCUs, unnecessary encoding processing increases because the unit of processing is 128×128 pixels at maximum, and the efficiency of encoding processing may decrease. For example, it may become difficult to ensure real-time operation of CABAC.

(18) Also, the coding formats suggested in NPL 2 and NPL 3 are not included in the AVC coding format, and compatibility with the I_PCM mode is not disclosed therein.

(19) The present disclosure has been made in view of these circumstances, and is directed to enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

Solution to Problem

(20) According to an aspect of the present disclosure, there is provided an image processing apparatus. The image processing apparatus includes an encoding mode setter that sets, in units of coding units having a hierarchical structure, whether a non-compression mode is to be selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and an encoder that encodes the image data in units of the coding units in accordance with a mode set by the encoding mode setter.

(21) The image processing apparatus may further include a shift processing controller that performs control, on a coding unit for which the non-compression mode has been set by the encoding mode setter, to skip shift processing in which a bit precision for encoding or decoding is increased; and a shift processor that performs the shift processing on a coding unit of the image data, the coding unit being controlled by the shift processing controller so as to undergo the shift processing.

(22) The image processing apparatus may further include a filter processing controller that performs control, on a coding unit for which the non-compression mode has been set by the encoding mode setter, to skip filter processing in which filtering is performed on a locally decoded image; a filter coefficient calculator that calculates a filter coefficient for the filter processing by using image data corresponding to a coding unit which is controlled by the filter processing controller so as to undergo the filter processing; and a filter processor that performs the filter processing in units of blocks, which are units of the filter processing, by using the filter coefficient calculated by the filter coefficient calculator.

(23) The filter processor may perform the filter processing on only pixels which are controlled by the filter processing controller so as to undergo the filter processing, the pixels being included in a current block which is a target to be processed.

(24) The image processing apparatus may further include a filter identification information generator that generates filter identification information in units of the blocks, the filter identification information being identification information indicating whether the filter processing is to be performed.

- (25) The filter processor may perform adaptive loop filtering on the locally decoded image, the adaptive loop filtering being adaptive filter processing using classification processing.
- (26) In a case where an amount of code of encoded data, which is obtained by encoding the image data corresponding to a current coding unit as a target of encoding processing, is smaller than or equal to an amount of input data, which is a data amount of the image data corresponding to the current coding unit, the encoding mode setter may set an encoding mode of the current coding unit to be the non-compression mode.
- (27) The image processing apparatus may further include an input data amount calculator that calculates the amount of input data. The encoding mode setter may compare, regarding the current coding unit, the amount of input data calculated by the input data amount calculator with the amount of code.
- (28) The image processing apparatus may further include an identification information generator that generates identification information in units of the coding units, the identification information indicating whether the non-compression mode has been set by the encoding mode setter.
- (29) According to an aspect of the present disclosure, there is provided an image processing method for an image processing apparatus. The image processing method includes setting, with an encoding mode setter, in units of coding units having a hierarchical structure, whether a non-compression mode is to be selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and encoding, with an encoder, the image data in units of the coding units in accordance with a set mode.
- (30) According to another aspect of the present disclosure, there is provided an image processing apparatus. The image processing apparatus includes an encoding mode determiner that determines, in units of coding units having a hierarchical structure, whether a non-compression mode has been selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and a decoder that decodes the encoding result in units of the coding units in accordance with a mode determined by the encoding mode determiner.
- (31) The image processing apparatus may further include a shift processing controller that performs control, on a coding unit for which the encoding mode determiner has determined that the non-compression mode has been selected, to skip shift processing in which a bit precision for encoding or decoding is increased; and a shift processor that performs the shift processing on a coding unit of the image data, the coding unit being controlled by the shift processing controller so as to undergo the shift processing.
- (32) The image processing apparatus may further include a filter processing controller that performs control, on a coding unit for which the encoding mode determiner has determined that the non-compression mode has been selected, to skip filter processing in which filtering is performed on a locally decoded image; and a filter processor that performs the filter processing on the image data in units of blocks, which are units of the filter processing. The filter processor may perform the filter processing on only pixels which have been controlled by the filter processing controller so as to undergo the filter processing, the pixels being included in a current block which is a target to be processed.
- (33) The filter processor may perform adaptive loop filtering on the locally decoded image, the adaptive loop filtering being adaptive filter processing using classification processing.
- (34) The filter processor may perform the filter processing, in a case where filter identification information indicating whether the filter processing has been performed indicates that the filter processing has been performed on image data corresponding to the current block which is a target to be processed, only when control is performed by the filter processing controller so as to perform the filter processing on all pixels included in the current block.
- (35) The encoding mode determiner may determine whether the non-compression mode has been

selected, on the basis of identification information indicating whether the non-compression mode has been selected in units of the coding units.

(36) According to another aspect of the present disclosure, there is provided an image processing method for an image processing apparatus. The image processing method includes determining, with an encoding mode determiner, in units of coding units having a hierarchical structure, whether a non-compression mode has been selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and decoding, with a decoder, the encoded data in units of the coding units in accordance with a determined mode.

(37) According to an aspect of the present disclosure, whether a non-compression mode is to be selected as an encoding mode for encoding image data is set in units of coding units having a hierarchical structure, the non-compression mode being an encoding mode in which the image data is output as encoded data, and the image data is encoded in units of the coding units in accordance with a set mode.

(38) According to another aspect of the present disclosure, whether a non-compression mode has been selected as an encoding mode for encoding image data is determined in units of coding units having a hierarchical structure, the non-compression mode being an encoding mode in which the image data is output as encoded data, and the encoded data is decoded in units of the coding units in accordance with a determined mode.

Advantageous Effects of Invention

(39) According to the present disclosure, an image can be processed. In particular, encoding efficiency can be enhanced while suppressing a decrease in the efficiency of encoding processing.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram illustrating an image encoding apparatus that outputs image compression information based on an AVC coding format.

(2) FIG. 2 is a block diagram illustrating an image decoding apparatus that receives image compression information based on the AVC coding format.

(3) FIG. 3 is a diagram illustrating an example of the type of macroblock.

(4) FIG. 4 is a diagram describing an example configuration of coding units.

(5) FIG. 5 is a diagram describing a method for increasing an amount of bits in internal arithmetic.

(6) FIG. 6 is a diagram describing an adaptive loop filter.

(7) FIG. 7 is a block diagram illustrating a main example configuration of an image encoding apparatus.

(8) FIG. 8 is a block diagram illustrating a main example configuration of the lossless encoder, loop filter, and PCM encoder in FIG. 7.

(9) FIG. 9 is a block diagram illustrating a main example configuration of the PCM deciding unit in FIG. 8.

(10) FIG. 10 is a flowchart describing an example of the flow of encoding processing.

(11) FIG. 11 is a flowchart describing an example of the flow of PCM encoding control processing.

(12) FIG. 12 is a flowchart describing an example of the flow of PCM encoding processing.

(13) FIG. 13 is a flowchart describing an example of the flow of reference image generation processing.

(14) FIG. 14 is a flowchart describing an example of the flow of loop filter processing.

(15) FIG. 15 is a block diagram illustrating a main example configuration of an image decoding apparatus.

(16) FIG. 16 is a block diagram illustrating a main example configuration of the lossless decoder,

loop filter, and PCM decoder in FIG. 15.

(17) FIG. 17 is a flowchart describing an example of the flow of decoding processing.

(18) FIG. 18 is a flowchart continued from FIG. 17, describing the example of the flow of decoding processing.

(19) FIG. 19 is a flowchart describing an example of the flow of loop filter processing.

(20) FIG. 20 is a diagram describing an example of I_PCM information.

(21) FIG. 21 is a block diagram illustrating a main example configuration of a personal computer.

(22) FIG. 22 is a block diagram illustrating a main example configuration of a television receiver.

(23) FIG. 23 is a block diagram illustrating a main example configuration of a mobile phone.

(24) FIG. 24 is a block diagram illustrating a main example configuration of a hard disk recorder.

(25) FIG. 25 is a block diagram illustrating a main example configuration of a camera.

DESCRIPTION OF EMBODIMENTS

(26) Hereinafter, embodiments for carrying out the present technology (hereinafter referred to as embodiments) will be described. Note that the description will be given in the following order. 1. First embodiment (image encoding apparatus) 2. Second embodiment (image decoding apparatus) 3. Third embodiment (personal computer) 4. Fourth embodiment (television receiver) 5. Fifth embodiment (mobile phone) 6. Sixth embodiment (hard disk recorder) 7. Seventh embodiment (camera)

1. First Embodiment

(27) [Image Encoding Apparatus Compatible with AVC Coding Format]

(28) FIG. 1 illustrates the configuration of an image encoding apparatus according to an embodiment, which encodes an image using an H.264 and MPEG (Moving Picture Experts Group) 4 Part 10 (AVC (Advanced Video Coding)) coding format.

(29) An image encoding apparatus **100** illustrated in FIG. 1 is an apparatus that encodes an image using a coding format based on the AVC standard and outputs the encoded image. As illustrated in FIG. 1, the image encoding apparatus **100** includes an A/D converter **101**, a screen rearrangement buffer **102**, a computing unit **103**, an orthogonal transform unit **104**, a quantizer **105**, a lossless encoder **106**, and an accumulation buffer **107**. Also, the image encoding apparatus **100** includes a dequantizer **108**, an inverse orthogonal transform unit **109**, a computing unit **110**, a deblocking filter **111**, a frame memory **112**, a selector **113**, an intra prediction unit **114**, a motion prediction/compensation unit **115**, a selector **116**, and a rate controller **117**.

(30) The A/D converter **101** A/D-converts image data input thereto, and outputs the image data to the screen rearrangement buffer **102** to store the image data therein. The screen rearrangement buffer **102** rearranges, in accordance with a GOP (Group of Picture) structure, frame images stored therein which are arranged in a display order so that the frame images are rearranged in an encoding order. The screen rearrangement buffer **102** supplies the rearranged frame images to the computing unit **103**. Also, the screen rearrangement buffer **102** supplies the rearranged frame images to the intra prediction unit **114** and the motion prediction/compensation unit **115**.

(31) The computing unit **103** subtracts, from an image read out from the screen rearrangement buffer **102**, a prediction image supplied from the intra prediction unit **114** or the motion prediction/compensation unit **115** via the selector **116**, and outputs difference information thereof to the orthogonal transform unit **104**.

(32) For example, in the case of an image on which intra coding is to be performed, the computing unit **103** subtracts a prediction image supplied from the intra prediction unit **114** from an image read out from the screen rearrangement buffer **102**. Also, for example, in the case of an image on which inter coding is to be performed, the computing unit **103** subtracts a prediction image supplied from the motion prediction/compensation unit **115** from an image read out from the screen rearrangement buffer **102**.

(33) The orthogonal transform unit **104** performs an orthogonal transform, such as the discrete cosine transform or Karhunen-Loeve transform, on the difference information supplied from the

computing unit **103**, and supplies a transform coefficient thereof to the quantizer **105**.

(34) The quantizer **105** quantizes the transform coefficient output from the orthogonal transform unit **104**. The quantizer **105** performs quantization by setting a quantization parameter on the basis of information about a target value of an amount of code supplied from the rate controller **117**. The quantizer **105** supplies the quantized transform coefficient to the lossless encoder **106**.

(35) The lossless encoder **106** performs lossless coding, such as variable-length coding or arithmetic coding, on the quantized transform coefficient. Coefficient data has been quantized under control performed by the rate controller **117**, and thus the amount of code thereof is equal to (or approximate to) a target value set by the rate controller **117**.

(36) The lossless encoder **106** obtains information indicating intra prediction and so forth from the intra prediction unit **114**, and obtains information indicating an inter prediction mode, motion vector information, and so forth from the motion prediction/compensation unit **115**. Note that information indicating intra prediction (intra-screen prediction) will also be referred to as intra prediction mode information hereinafter. Also, information indicating an information mode indicating inter prediction (inter-screen prediction) will also be referred to as inter prediction mode information.

(37) The lossless encoder **106** encodes the quantized transform coefficient, and also causes various pieces of information, such as a filter coefficient, intra prediction mode information, inter prediction mode information, and a quantization parameter, to be part of header information of encoded data (multiplexes the various pieces of information). The lossless encoder **106** supplies encoded data which has been obtained through encoding to the accumulation buffer **107** to store the encoded data therein.

(38) For example, in the lossless encoder **106**, lossless coding processing such as variable-length coding or arithmetic coding is performed. An example of variable-length coding includes CAVLC (Context-Adaptive Variable Length Coding) defined in the H.264/AVC format. An example of arithmetic coding includes CABAC (Context-Adaptive Binary Arithmetic Coding).

(39) The accumulation buffer **107** temporarily holds the encoded data supplied from the lossless encoder **106**, and outputs the encoded data to, for example, a recording device or a transmission channel in the subsequent stage (not illustrated) at a certain timing, as an encoded image which has been encoded using the H.264/AVC format.

(40) In addition, the transform coefficient quantized by the quantizer **105** is also supplied to the dequantizer **108**. The dequantizer **108** dequantizes the quantized transform coefficient using a method corresponding to the quantization performed by the quantizer **105**. The dequantizer **108** supplies the transform coefficient obtained thereby to the inverse orthogonal transform unit **109**.

(41) The inverse orthogonal transform unit **109** performs inverse orthogonal transform on the transform coefficient supplied thereto using a method corresponding to the orthogonal transform processing performed by the orthogonal transform unit **104**. The output obtained through the inverse orthogonal transform (recovered difference information) is supplied to the computing unit **110**.

(42) The computing unit **110** adds a prediction image supplied from the intra prediction unit **114** or the motion prediction/compensation unit **115** via the selector **116** to the result of inverse orthogonal transform supplied from the inverse orthogonal transform unit **109**, that is, recovered difference information, and obtains a locally decoded image (decoded image).

(43) For example, in a case where the difference information corresponds to an image on which intra coding is to be performed, the computing unit **110** adds a prediction image supplied from the intra prediction unit **114** to the difference information. Also, for example, in a case where the difference information corresponds to an image on which inter coding is to be performed, the computing unit **110** adds a prediction image supplied from the motion prediction/compensation unit **115** to the difference information.

(44) The result of addition is supplied to the deblocking filter **111** or the frame memory **112**.

- (45) The deblocking filter **111** performs deblocking filter processing as appropriate, thereby removing a block distortion of a decoded image. The deblocking filter **111** supplies the result of filter processing to the frame memory **112**. Note that the decoded image output from the computing unit **110** can be supplied to the frame memory **112** without via the deblocking filter **111**. That is, deblocking filter processing by the deblocking filter **111** can be skipped.
- (46) The frame memory **112** stores the decoded image supplied thereto, and outputs the stored decoded image as a reference image to the intra prediction unit **114** or the motion prediction/compensation unit **115** via the selector **113** at a certain timing.
- (47) For example, in the case of an image on which intra coding is to be performed, the frame memory **112** supplies a reference image to the intra prediction unit **114** via the selector **113**. Also, for example, in a case where inter coding is to be performed, the frame memory **112** supplies a reference image to the motion prediction/compensation unit **115** via the selector **113**.
- (48) In a case where the reference image supplied from the frame memory **112** is an image on which intra coding is to be performed, the selector **113** supplies the reference image to the intra prediction unit **114**. On the other hand, in a case where the reference image supplied from the frame memory **112** is an image on which inter coding is to be performed, the selector **113** supplies the reference image to the motion prediction/compensation unit **115**.
- (49) The intra prediction unit **114** performs intra prediction (intra-screen prediction) in which a prediction image is generated using pixel values of a target picture to be processed supplied from the frame memory **112** via the selector **113**. The intra prediction unit **114** performs intra prediction using a plurality of prepared modes (intra prediction modes).
- (50) In the H.264 image information coding format, an intra 4×4 prediction mode, an intra 8×8 prediction mode, and an intra 16×16 prediction mode are defined for luminance signals. Also, regarding color-difference signals, a prediction mode independent of that for luminance signals can be defined for individual macroblocks. Regarding the intra 4×4 prediction mode, one intra prediction mode is defined for each 4×4 luminance block. Regarding the intra 8×8 prediction mode, one intra prediction mode is defined for each 8×8 luminance block. Regarding the intra 16×16 prediction mode and color-difference signals, one prediction mode is defined for one macroblock.
- (51) The intra prediction unit **114** generates prediction images using all candidate intra prediction modes, evaluates the cost function values of the individual prediction images using an input image supplied the screen rearrangement buffer **102**, and selects an optimal mode. After selecting an optimal intra prediction mode, the intra prediction unit **114** supplies the prediction image which has been generated using the optimal mode to the computing unit **103** and the computing unit **110** via the selector **116**.
- (52) Also, as described above, the intra prediction unit **114** supplies information, such as intra prediction mode information indicating the adopted intra prediction mode, to the lossless encoder **106** as appropriate.
- (53) The motion prediction/compensation unit **115** performs motion prediction (inter prediction) on an image on which inter coding is to be performed, using an input image supplied from the screen rearrangement buffer **102** and a reference image supplied from the frame memory **112** via the selector **113**, performs motion compensation processing in accordance with a detected motion vector, and generates a prediction image (inter prediction image information). The motion prediction/compensation unit **115** performs such inter prediction using a plurality of prepared modes (inter prediction modes).
- (54) The motion prediction/compensation unit **115** generates prediction images using all candidate inter prediction modes, evaluates the cost function values of the individual prediction images, and selects an optimal mode. The motion prediction/compensation unit **115** supplies the generated prediction image to the computing unit **103** and the computing unit **110** via the selector **116**.
- (55) Also, the motion prediction/compensation unit **115** supplies inter prediction mode information

indicating the adopted inter prediction mode and motion vector information indicating a calculated motion vector to the lossless encoder **106**.

(56) In the case of an image on which intra coding is to be performed, the selector **116** supplies the output of the intra prediction unit **114** to the computing unit **103** and the computing unit **110**. In the case of an image on which inter coding is to be performed, the selector **116** supplies the output of the motion prediction/compensation unit **115** to the computing unit **103** and the computing unit **110**.

(57) The rate controller **117** controls the rate of the quantization operation performed by the quantizer **105** on the basis of the compressed images accumulated in the accumulation buffer **107** so that overflow or underflow does not occur.

(58) [Image Decoding Apparatus Compatible with AVC Coding Format]

(59) FIG. 2 is a block diagram illustrating a main example configuration of an image decoding apparatus that realizes image compression using an orthogonal transform, such as the discrete cosine transform or Karhunen-Loeve transform, and by motion compensation. The image decoding apparatus **200** illustrated in FIG. 2 is a decoding apparatus corresponding to the image encoding apparatus **100** illustrated in FIG. 1.

(60) Encoded data which has been encoded by the image encoding apparatus **100** is supplied to the image decoding apparatus **200** corresponding to the image encoding apparatus **100** via an arbitrary path, for example, a transmission channel, a recording medium, or the like, and is decoded.

(61) As illustrated in FIG. 2, the image decoding apparatus **200** includes an accumulation buffer **201**, a lossless decoder **202**, a dequantizer **203**, an inverse orthogonal transform unit **204**, a computing unit **205**, a deblocking filter **206**, a screen rearrangement buffer **207**, and a D/A converter **208**. Also, the image decoding apparatus **200** includes a frame memory **209**, a selector **210**, an intra prediction unit **211**, a motion prediction/compensation unit **212**, and a selector **213**.

(62) The accumulation buffer **201** accumulates encoded data transmitted thereto. The encoded data has been encoded by the image encoding apparatus **100**. The lossless decoder **202** decodes encoded data read out from the accumulation buffer **201** at a certain timing, using a format corresponding to the coding format used by the lossless encoder **106** illustrated in FIG. 1.

(63) Also, in a case where a target frame has been intra coded, intra prediction mode information is stored in the header portion of the encoded data. The lossless decoder **202** also decodes the intra prediction mode information, and supplies the information to the intra prediction unit **211**. In contrast, in a case where a target frame has been inter coded, motion vector information is stored in the header portion of the encoded data. The lossless decoder **202** also decodes the motion vector information, and supplies the information to the motion prediction/compensation unit **212**.

(64) The dequantizer **203** dequantizes coefficient data (quantized coefficient) which is obtained through decoding performed by the lossless decoder **202**, using a method corresponding to the quantization method used by the quantizer **105** illustrated in FIG. 1. That is, the dequantizer **203** dequantizes the quantized coefficient using a method similar to that used by the dequantizer **108** illustrated in FIG. 1.

(65) The dequantizer **203** supplies the dequantized coefficient data, that is, an orthogonal transform coefficient, to the inverse orthogonal transform unit **204**. The inverse orthogonal transform unit **204** performs inverse orthogonal transform on the orthogonal transform coefficient using a method corresponding to the orthogonal transform method used by the orthogonal transform unit **104** illustrated in FIG. 1 (a method similar to the method used by the inverse orthogonal transform unit **109** illustrated in FIG. 1), and obtains decoded residual data corresponding to residual data before orthogonal transform is performed by the image encoding apparatus **100**. For example, fourth-order inverse orthogonal transform is performed.

(66) The decoded residual data obtained through inverse orthogonal transform is supplied to the computing unit **205**. Also, the computing unit **205** is supplied with a prediction image from the intra prediction unit **211** or the motion prediction/compensation unit **212** via the selector **213**.

(67) The computing unit **205** adds the decoded residual data and the prediction image, thereby obtaining decoded image data corresponding to the image data from which the prediction image has not been subtracted by the computing unit **103** of the image encoding apparatus **100**. The computing unit **205** supplies the decoded image data to the deblocking filter **206**.

(68) The deblocking filter **206** removes a block distortion of the decoded image supplied thereto, and supplies the decoded image to the screen rearrangement buffer **207**.

(69) The screen rearrangement buffer **207** rearranges images. That is, the frames which have been rearranged in an encoding order by the screen rearrangement buffer **102** illustrated in FIG. **1** are rearranged in the original display order. The D/A converter **208** D/A-converts the image supplied from the screen rearrangement buffer **207**, and outputs the image to a display (not illustrates) to display the image thereon.

(70) The output of the deblocking filter **206** is also supplied to the frame memory **209**.

(71) The frame memory **209**, the selector **210**, the intra prediction unit **211**, the motion prediction/compensation unit **212**, and the selector **213** respectively correspond to the frame memory **112**, the selector **113**, the intra prediction unit **114**, the motion prediction/compensation unit **115**, and the selector **116** of the image encoding apparatus **100**.

(72) The selector **210** reads out an image on which inter processing is to be performed and a reference image from the frame memory **209**, and supplies the images to the motion prediction/compensation unit **212**. Also, the selector **210** reads out an image to be used for intra prediction from the frame memory **209**, and supplies the image to the intra prediction unit **211**.

(73) The intra prediction unit **211** is supplied with, for example, information indicating an intra prediction mode which has been obtained by decoding header information from the lossless decoder **202** as appropriate. The intra prediction unit **211** generates a prediction image from the reference image obtained from the frame memory **209** on the basis of the information, and supplies the generated prediction image to the selector **213**.

(74) The motion prediction/compensation unit **212** obtains information which has been obtained by decoding the header information (prediction mode information, motion vector information, reference frame information, flag, various parameters, etc.) from the lossless decoder **202**.

(75) The motion prediction/compensation unit **212** generates a prediction image from the reference image obtained from the frame memory **209** on the basis of the information supplied from the lossless decoder **202**, and supplies the generated prediction image to the selector **213**.

(76) The selector **213** selects the prediction image which has been generated by the motion prediction/compensation unit **212** or the intra prediction unit **211**, and supplies the image to the computing unit **205**.

(77) [Macroblock Type]

(78) By the way, as disclosed in PTL 1, there is a coding format in which an encoding mode for encoding and outputting image data and a non-encoding mode for outputting image data without encoding the image data are provided, whether the encoding mode or the non-encoding mode is to be used is selected in units of macroblocks, and the encoding mode and the non-encoding mode can be used in combination within a single picture. As disclosed in PTL 2, also in the AVC coding format, an I_PCM (Intra-block pulse code modulation) mode (non-compression mode) for outputting image data without encoding the image data is supported as one of the types of macroblocks (mb_type), as illustrated in FIG. **3**.

(79) This is used for ensuring real-time operation of arithmetic coding processing in a case where a quantization parameter is set to be a small value, such as QP=0, and in a case where the information amount of encoded data is larger than that of an input image. Also, lossless coding can be realized by using the I-PCM mode (non-compression mode).

(80) [Cost Function]

(81) Meanwhile, to achieve higher encoding efficiency in the AVC coding format, it is important to select an appropriate prediction mode.

(82) An example of a selection method is a method which is loaded in reference software of H.264/MPEG-4 AVC called JM (Joint Model), which is released in <http://iphome.hhi.de/suehring/tml/index.htm>.

(83) According to JM, the following two mode determination methods, that is, High Complexity Mode and Low Complexity Mode, can be selected. In either of them, cost function values regarding individual prediction modes are calculated, and a prediction mode with the smallest cost function value is selected as an optimal mode for a target block or macroblock.

(84) The cost function regarding the High Complexity Mode is expressed by the following equation (1).

$$\text{Cost}(\text{Mode} \in \Omega) = D + \lambda \star R \quad (1)$$

(86) Here, Ω represents a universal set of candidate modes for encoding a target block or macroblock, and D represents the differential energy between a decoded image and an input image in a case where encoding is performed using the prediction mode. λ represents a Lagrange undetermined multiplier which is given as a function of a quantization parameter. R represents a total amount of code including an orthogonal transform coefficient in a case where encoding is performed using the mode.

(87) That is, to perform encoding using the High Complexity Mode, it is necessary to once perform preliminary encoding processing using all the candidate modes in order to calculate the parameters D and R , which involves a larger amount of computation.

(88) The cost function in the Low Complexity Mode is expressed by the following equation (2).

$$\text{Cost}(\text{Mode} \in \Omega) = D + \text{QP2Quant}(\text{QP}) \star \text{HeaderBit} \quad (2)$$

(90) Here, D represents the differential energy between a prediction image and an input image, unlike in the High Complexity Mode. $\text{QP2Quant}(\text{QP})$ is given as a function of a quantization parameter QP , and HeaderBit represents an amount of code regarding information belonging to the Header, such as a motion vector and mode, not including an orthogonal transform coefficient.

(91) That is, in the Low Complexity Mode, it is necessary to perform prediction processing regarding individual candidate modes, but a decoded image is not necessary, and thus it is not necessary to perform encoding processing. Thus, the Low Complexity Mode can be realized with an amount of computation smaller than that of the High Complexity Mode.

(92) [Coding Unit]

(93) Next, description will be given of coding units, which are defined in the HEVC coding format described in NPL 1.

(94) Coding units (CUs) are also called coding tree blocks (CTBs), are partial regions of an image of each picture which play a role similar to macroblocks in AVC, and are coding units having a hierarchical structure. The size of a macroblock is fixed to 16×16 pixels, whereas the size of a CU is not fixed and is specified in image compression information in each sequence.

(95) In particular, a CU having the largest size is called an LCU (Largest Coding Unit), and a CU having the smallest size is called an SCU (Smallest Coding Unit). For example, the sizes of these regions are specified in a sequence parameter set (SPS) included in image compression information. The individual regions are square shaped, and the sizes thereof are limited to sizes expressed by a power of 2.

(96) FIG. 4 illustrates an example of coding units defined in HEVC. In the example illustrated in FIG. 4, the size of LCU is 128, and the maximum layer depth is 5. When the value of `split_flag` is "1", the CU having a size of $2N \times 2N$ is split into CUs each having a size of $N \times N$ in the immediately lower layer.

(97) Furthermore, a CU is split into prediction units (PUs), each being a region serving as a unit of processing for intra or inter prediction (partial region of an image of each picture), or is split into transform units (TUs), each being a region serving as a unit of processing for orthogonal transform (partial region of an image of each picture). At present, in HEVC, 16×16 and 32×32 orthogonal

transform can be used in addition to 4×4 and 8×8 orthogonal transform.

(98) [IBDI]

(99) Meanwhile, NPL 2 suggests a method for increasing internal arithmetic (IBDI (Internal bit depth increase except frame memory)) illustrated in FIG. 5. In this method, as illustrated in FIG. 5, the bit depth of data is increased, for example, from 8 bits to 12 bits, in quantization processing, lossless coding processing, dequantization processing, filter processing, prediction processing, lossless decoding processing, and so forth performed by an encoding apparatus and a decoding apparatus. Accordingly, an internal arithmetic error in processing such as orthogonal transform or motion compensation can be decreased, and encoding efficiency can be enhanced.

(100) [BALF]

(101) Meanwhile, NPL 3 suggests a method in which an FIR filter is provided in a motion compensation loop and loop filter processing using the filter (BALF (Block-based Adaptive Loop Filter)) is adaptively performed, as illustrated in FIG. 5. In an encoding apparatus, the FIR filter coefficient is obtained using a Wiener filter so as to minimize an error with respect to an input image, and thereby degradation in a reference image can be minimized, and encoding efficiency of image compression information to be output can be enhanced.

(102) [Efficiency of Encoding Processing]

(103) Meanwhile, in the case of a coding format in which CUs are defined and various processing operations are performed in units of CUs, as in HEVC, it can be considered that a macroblock in AVC corresponds to an LCU. However, since CUs have a hierarchical structure as illustrated in FIG. 4, the size of the LCU in the top layer is generally set to be larger than a macroblock in AVC, for example, 128×128 pixels.

(104) Therefore, in such a coding format, as in the case of AVC, if the I_PCM mode is set in units of LCUs, the unit of processing becomes larger than that in AVC, for example, 128×128 pixels.

(105) The mode of intra prediction or inter prediction is determined by calculating and comparing cost function values, as described above. That is, prediction and encoding are performed using all the modes, individual cost function values are calculated, an optimal mode is selected, and encoded data is generated using the optimal mode.

(106) However, when the I_PCM mode is adopted, the encoded data generated using the optimal mode is discarded, and an input image (non-encoded data) is adopted as an encoding result. Thus, when the I_PCM mode is selected, all the processing operations for generating the encoded data of the optimal mode are not necessary. That is, if the unit of selection control of the I_PCM mode becomes large, unnecessary processing operations further increase. That is, as described above, if it is selected for each LCU whether or not the I_PCM mode is to be adopted, the efficiency of encoding processing may further decrease. Thus, for example, it may become difficult to ensure real-time operation of CABAC.

(107) Also, the above-described technologies such as IBDI and BALF are not included in the AVC coding format. In a case where the I_PCM mode is adopted, it is unknown how to control these processing operations.

(108) Accordingly, the present embodiment enables more detailed control of selection of the I_PCM mode (non-compression mode), and also enables enhancement of encoding efficiency while suppressing a decrease in the efficiency of encoding processing. Also, the present embodiment enables appropriate control of execution of IBDI and BALF according to selection of the I_PCM mode, and also enables further suppression of a decrease in the efficiency of encoding processing.

(109) [Image Encoding Apparatus]

(110) FIG. 7 is a block diagram illustrating a main example configuration of an image encoding apparatus.

(111) The image encoding apparatus **300** illustrated in FIG. 7 is basically similar to the image encoding apparatus **100** illustrated in FIG. 1, and encodes image data. As illustrated in FIG. 7, the

image encoding apparatus **300** includes an A/D converter **301**, a screen rearrangement buffer **302**, an adaptive shift-to-left unit **303**, a computing unit **304**, an orthogonal transform unit **305**, a quantizer **306**, a lossless encoder **307**, and an accumulation buffer **308**. Also, the image encoding apparatus **300** includes a dequantizer **309**, an inverse orthogonal transform unit **310**, a computing unit **311**, a loop filter **312**, an adaptive shift-to-right unit **313**, a frame memory **314**, an adaptive shift-to-left unit **315**, a selector **316**, an intra prediction unit **317**, a motion prediction/compensation unit **318**, a selector **319**, and a rate controller **320**.

(112) The image encoding apparatus **300** further includes a PCM encoder **321**.

(113) The A/D converter **301** A/D-converts image data input thereto, as in the case of the A/D converter **101**. The A/D converter **301** supplies the converted image data (digital data) to the screen rearrangement buffer **302** to store the image data therein. The screen rearrangement buffer **302** rearranges, in accordance with a GOP (Group of Picture) structure, frame images stored therein which are arranged in a display order so that the frame images are rearranged in an encoding order, as in the case of the screen rearrangement buffer **102**. The screen rearrangement buffer **302** supplies the rearranged frame images to the adaptive shift-to-left unit **303**.

(114) In addition, the screen rearrangement buffer **302** supplies the rearranged frame images also to the lossless encoder **307** and the PCM encoder **321**.

(115) The adaptive shift-to-left unit **303** is controlled by the PCM encoder **321**, shifts image data which has been read out from the screen rearrangement buffer **302** in the left direction, and increases the bit depth thereof by a certain number of bits (for example, 4 bits). For example, the adaptive shift-to-left unit **303** increases the bit depth of the image data which has been read out from the screen rearrangement buffer **302** from 8 bits to 12 bits. As a result of increasing the bit depth in this way, the accuracy of internal arithmetic in each of processing operations, such as orthogonal transform processing, quantization processing, lossless coding processing, prediction processing, and so forth can be increased, and errors can be suppressed.

(116) Note that the amount of shift to the left (the amount of bits) is not specified, and may be fixed or variable. Also, the shift-to-left processing may be skipped in accordance with control performed by the PCM encoder **321**.

(117) The adaptive shift-to-left unit **303** supplies the image data on which shift-to-left processing has been performed to the computing unit **304** (in a case where the processing is skipped, the image data output from the screen rearrangement buffer **302** is supplied to the computing unit **304**). In addition, the adaptive shift-to-left unit **303** also supplies the image data to the intra prediction unit **317** and the motion prediction/compensation unit **318**.

(118) The computing unit **304** subtracts, from the image supplied from the adaptive shift-to-left unit **303**, a prediction image supplied from the intra prediction unit **317** or the motion prediction/compensation unit **318** via the selector **319**, as in the case of the computing unit **103**. The computing unit **304** outputs difference information thereof to the orthogonal transform unit **305**.

(119) For example, in the case of an image on which intra coding is to be performed, the computing unit **304** subtracts a prediction image supplied from the intra prediction unit **317** from the image supplied from the adaptive shift-to-left unit **303**. Also, for example, in the case of an image on which inter coding is to be performed, the computing unit **304** subtracts a prediction image supplied from the motion prediction/compensation unit **318** from the image supplied from the adaptive shift-to-left unit **303**.

(120) The orthogonal transform unit **305** performs an orthogonal transform, such as the discrete cosine transform or Karhunen-Loeve transform, on the difference information supplied from the computing unit **304**, as in the case of the orthogonal transform unit **104**. The method for the orthogonal transform is not specified. The orthogonal transform unit **305** supplies a transform coefficient thereof to the quantizer **306**.

(121) The quantizer **306** quantizes the transform coefficient supplied from the orthogonal transform

unit **305**, as in the case of the quantizer **105**. The quantizer **306** performs quantization by setting a quantization parameter on the basis of information about a target value of an amount of code supplied from the rate controller **320**. The method for the quantization is not specified. The quantizer **306** supplies the quantized transform coefficient to the lossless encoder **307**.

(122) The lossless encoder **307** performs lossless coding, such as variable-length coding or arithmetic coding, on the transform coefficient quantized by the quantizer **306**, as in the case of the lossless encoder **106**. Coefficient data has been quantized under control performed by the rate controller **320**, and thus the amount of code thereof is equal to (or approximate to) a target value set by the rate controller **320**.

(123) Note that, in a case where the I_PCM mode is selected by the PCM encoder **321**, the lossless encoder **307** regards an input image (non-encoded data) supplied from the screen rearrangement buffer **302** as an encoding result (that is, encoding is actually skipped).

(124) Also, the lossless encoder **307** obtains information indicating the mode of intra prediction and so forth from the intra prediction unit **317**, and obtains information indicating the mode of inter prediction, motion vector information, and so forth from the motion prediction/compensation unit **318**. Furthermore, the lossless encoder **307** obtains the filter coefficient used by the loop filter **312**.

(125) The lossless encoder **307** encodes the various pieces of information, such as a filter coefficient, information indicating the mode of intra prediction mode or inter prediction, and a quantization parameter, as in the case of the lossless encoder **106**, and causes the various pieces of information to be part of header information of encoded data (multiplexes the various pieces of information). The lossless encoder **307** supplies encoded data which has been obtained through encoding (including non-encoded data in the case of the I_PCM mode) to the accumulation buffer **308** to store the encoded data therein.

(126) For example, in the lossless encoder **307**, as in the case of the lossless encoder **106**, lossless coding processing such as variable-length coding or arithmetic coding is performed. An example of variable-length coding includes CAVLC (Context-Adaptive Variable Length Coding) defined in the H.264/AVC format. An example of arithmetic coding includes CABAC (Context-Adaptive Binary Arithmetic Coding). Of course, the lossless encoder **307** may perform encoding using a method other than these methods.

(127) The accumulation buffer **308** temporarily holds the encoded data supplied from the lossless encoder **307** (including non-encoded data in the case of the I_PCM mode), as in the case of the accumulation buffer **107**. The accumulation buffer **308** outputs the encoded data held therein to, for example, a recording device (recording medium) or a transmission channel in the subsequent stage (not illustrated) at a certain timing.

(128) In addition, the transform coefficient quantized by the quantizer **306** is also supplied to the dequantizer **309**. The dequantizer **309** dequantizes the quantized transform coefficient using a method corresponding to the quantization performed by the quantizer **306**, as in the case of the dequantizer **108**. The method for the dequantization is not limited as long as the method corresponds to the quantization processing performed by the quantizer **306**. The dequantizer **309** supplies the transform coefficient obtained thereby to the inverse orthogonal transform unit **310**.

(129) The inverse orthogonal transform unit **310** performs inverse orthogonal transform on the transform coefficient supplied from the dequantizer **309** using a method corresponding to the orthogonal transform processing performed by the orthogonal transform unit **305**, as in the case of the inverse orthogonal transform unit **109**. The method for the inverse orthogonal transform is not limited as long as the method corresponds to the orthogonal transform processing performed by the orthogonal transform unit **305**. The output obtained through the inverse orthogonal transform (recovered difference information) is supplied to the computing unit **311**.

(130) The computing unit **311** adds a prediction image supplied from the intra prediction unit **317** or the motion prediction/compensation unit **318** via the selector **319** to the result of inverse orthogonal transform supplied from the inverse orthogonal transform unit **310**, that is, recovered

difference information, and obtains a locally decoded image (decoded image), as in the case of the computing unit **110**.

(131) For example, in a case where the difference information corresponds to an image on which intra coding is to be performed, the computing unit **311** adds a prediction image supplied from the intra prediction unit **317** to the difference information. Also, for example, in a case where the difference information corresponds to an image on which inter coding is to be performed, the computing unit **311** adds a prediction image supplied from the motion prediction/compensation unit **318** to the difference information.

(132) The result of addition (decoded image) is supplied to the loop filter **312** or the adaptive shift-to-right unit **313**.

(133) The loop filter **312** includes a deblocking filter, an adaptive loop filter, or the like, and performs filter processing on the decoded image supplied from the computing unit **311** as appropriate. For example, the loop filter **312** performs deblocking filter processing which is similar to that performed by the deblocking filter **111** on the decoded image, thereby removing a block distortion of the decoded image. Also, for example, the loop filter **312** is controlled by the PCM encoder **321**, and performs loop filter processing on the result of the deblocking filter processing (the decoded image from which a block distortion has been removed) by using a Wiener filter, thereby improving the image quality. Note that the adaptive loop filter processing may be skipped in accordance with control performed by the PCM encoder **321**.

(134) Alternatively, the loop filter **312** may perform arbitrary filter processing on the decoded image. Also, the loop filter **312** may supply a filter coefficient used for filter processing to the lossless encoder **307** as necessary, so that the filter coefficient is encoded.

(135) The loop filter **312** supplies the result of filter processing (the decoded image on which filter processing has been performed) to the adaptive shift-to-right unit **313**. Note that, as described above, the decoded image output from the computing unit **311** may be supplied to the adaptive shift-to-right unit **313** without via the loop filter **312**. That is, filter processing by the loop filter **312** may be skipped.

(136) The adaptive shift-to-right unit **313** is controlled by the PCM encoder **321**, shifts the image data supplied from the computing unit **311** or the loop filter **312** in the right direction, and decreases the bit depth thereof by a certain number of bits (for example, 4 bits). That is, the adaptive shift-to-right unit **313** shifts, to the right, the image data by the number of bits by which the image data has been shifted to the left by the adaptive shift-to-left unit **303**, so as to change the bit depth of the image data to the state before the image data is shifted to the left (the state at the time when the image data is read out from the screen rearrangement buffer **302**).

(137) For example, the adaptive shift-to-right unit **313** decreases the bit depth of the image data supplied from the computing unit **311** or the loop filter **312** from 12 bits to 8 bits. As a result of decreasing the bit depth in this way, the data amount of image data stored in the frame memory can be decreased.

(138) Note that the amount of shift to the right (the amount of bits) is not specified as long as the amount matches the amount of shift to the left in the adaptive shift-to-left unit **303**. That is, the amount may be fixed or variable. Also, the shift-to-right processing may be skipped in accordance with control performed by the PCM encoder **321**.

(139) The adaptive shift-to-right unit **313** supplies the image data on which shift-to-right processing has been performed to the frame memory **314** (in a case where the processing is skipped, the image data output from the computing unit **311** or the loop filter **312** is supplied to the frame memory **314**).

(140) The frame memory **314** stores the decoded image supplied thereto, as in the case of the frame memory **112**, and outputs the stored decoded image to the adaptive shift-to-left unit **315** as a reference image at a certain timing.

(141) The adaptive shift-to-left unit **315** is a processing unit similar to the adaptive shift-to-left unit

303, is controlled by the PCM encoder **321**, appropriately shifts the image data (reference image) read out from the frame memory **314** in the left direction, and increases the bit depth thereof by a certain number of bits (for example, 4 bits).

(142) For example, in a case where the mode is not the I_PCM mode, the data of an input image is shifted to the left by the adaptive shift-to-left unit **303**. Thus, the adaptive shift-to-left unit **315** shifts the data of a reference image read out from the frame memory **314** to the left in accordance with control performed by the PCM encoder **321**, and increases the bit depth by the number of bits that is the same as in the case of the adaptive shift-to-left unit **303** (for example, changes the bit depth from 8 bits to 12 bits).

(143) Then, the adaptive shift-to-left unit **315** supplies the image data on which shift-to-left processing has been performed to the selector **316**. As a result of increasing the bit depth in this way, the bit depth of the reference image can be made the same as the bit depth of the input image, and the reference image can be added to the input image. Also, the accuracy of internal arithmetic such as prediction processing can be increased, and errors can be suppressed.

(144) In contrast, for example, in the case of the I_PCM mode, the data of an input image is not shifted to the left by the adaptive shift-to-left unit **303**. Thus, the adaptive shift-to-left unit **315** supplies the reference image read out from the frame memory **314** to the selector **316** without increasing the bit depth, in accordance with control performed by the PCM encoder **321**.

(145) In the case of intra prediction, the selector **316** supplies the reference image supplied from the adaptive shift-to-left unit **315** to the intra prediction unit **317**, as in the case of the selector **113**. Also, in the case of inter prediction, the selector **316** supplies the reference image supplied from the adaptive shift-to-left unit **315** to the motion prediction/compensation unit **318**, as in the case of the selector **113**.

(146) The intra prediction unit **317** performs intra prediction (intra-screen prediction) in which a prediction image is generated using the reference image supplied from the adaptive shift-to-left unit **315** via the selector **316**. The intra prediction unit **317** performs intra prediction using a plurality of prepared modes (intra prediction modes). The intra prediction unit **317** is also capable of performing intra prediction using an arbitrary mode other than the modes defined in the AVC coding format.

(147) The intra prediction unit **317** generates prediction images using all candidate intra prediction modes, evaluates the cost function values of the individual prediction images using an input image supplied the adaptive shift-to-left unit **303**, and selects an optimal mode. After selecting an optimal intra prediction mode, the intra prediction unit **317** supplies the prediction image which has been generated using the optimal mode to the computing unit **304** and the computing unit **311** via the selector **319**.

(148) Also, as described above, the intra prediction unit **317** supplies information, such as intra prediction mode information indicating the adopted intra prediction mode, to the lossless encoder **307** as appropriate, so that the information is encoded.

(149) The motion prediction/compensation unit **318** performs motion prediction (inter prediction) on an image on which inter coding is to be performed, using an input image supplied from the adaptive shift-to-left unit **303** and a reference image supplied from the adaptive shift-to-left unit **315** via the selector **316**, performs motion compensation processing in accordance with a detected motion vector, and generates a prediction image (inter prediction image information). The motion prediction/compensation unit **318** performs such inter prediction using a plurality of prepared modes (inter prediction modes). The motion prediction/compensation unit **318** is also capable of performing inter prediction using an arbitrary mode other than the modes defined in the AVC coding format.

(150) The motion prediction/compensation unit **318** generates prediction images using all candidate inter prediction modes, evaluates the cost function values of the individual prediction images, and selects an optimal mode. After selecting the optimal inter prediction mode, the motion

prediction/compensation unit **318** supplies the prediction image generated using the optimal mode to the computing unit **304** and the computing unit **311** via the selector **319**.

(151) Also, the motion prediction/compensation unit **318** supplies inter prediction mode information indicating the adopted inter prediction mode and motion vector information indicating a calculated motion vector to the lossless encoder **307**, so that the information is encoded.

(152) In the case of an image on which intra coding is to be performed, the selector **319** supplies the output of the intra prediction unit **317** to the computing unit **304** and the computing unit **311**, as in the case of the selector **116**. In the case of an image on which inter coding is to be performed, the selector **319** supplies the output of the motion prediction/compensation unit **318** to the computing unit **304** and the computing unit **311**.

(153) The rate controller **320** controls the rate of the quantization operation performed by the quantizer **306** on the basis of the amount of code of the encoded data accumulated in the accumulation buffer **308** so that overflow or underflow does not occur.

(154) Also, the rate controller **320** supplies the amount of code (the amount of generated code) of the encoded data accumulated in the accumulation buffer **308** to the PCM encoder **321**.

(155) The PCM encoder **321** compares the amount of code supplied from the rate controller **320** with the data amount of the input image supplied from the screen rearrangement buffer **302**, and selects whether or not the I_PCM mode is to be adopted. At this time, the PCM encoder **321** performs selection in units of CUs, which are smaller than LCUs. That is, the PCM encoder **321** controls whether or not the I_PCM mode is to be adopted in more detail.

(156) In accordance with the result of selection, the PCM encoder **321** controls the operations of the lossless encoder **307**, the adaptive shift-to-left unit **303**, the adaptive shift-to-right unit **313**, the adaptive shift-to-left unit **315**, and the loop filter **312**.

(157) [Lossless Encoder, PCM Encoder, and Loop Filter]

(158) FIG. **8** is a block diagram illustrating a main example configuration of the lossless encoder **307**, the PCM encoder **321**, and the loop filter **312** illustrated in FIG. **7**.

(159) As illustrated in FIG. **8**, the lossless encoder **307** includes a NAL (Network Abstraction Layer) encoder **331** and a CU encoder **332**.

(160) The NAL encoder **331** encodes a NAL, for example, an SPS (Sequence Parameter Set), a PPS (Picture Parameter Set), or the like, on the basis of a user instruction, specifications, or the like input via a user interface (not illustrated). The NAL encoder **331** supplies the encoded NAL (NAL data) to the accumulation buffer **308**, so that the NAL data is added to CU data, which is an encoded VCL (Video Coding Layer) supplied from the CU encoder **332** to the accumulation buffer **308**.

(161) The CU encoder **332** is controlled by the PCM encoder **321** (on the basis of an On/Off control signal supplied from the PCM encoder **321**) and encodes a VCL. For example, in a case where the I_PCM mode is not selected by the PCM encoder **321** (in a case where a control signal representing “On” is supplied from the PCM encoder **321**), the CU encoder **332** encodes quantized orthogonal transform coefficients of individual CUs. The CU encoder **332** supplies the pieces of encoded data (CU data) of the individual CUs to the accumulation buffer **308**.

(162) Also, for example, in a case where the I_PCM mode is selected by the PCM encoder **321** (in a case where a control signal representing “Off” is supplied from the PCM encoder **321**), the CU encoder **332** supplies input pixel values which are supplied from the screen rearrangement buffer **302** to the accumulation buffer **308**, as an encoding result (CU data).

(163) In addition, the CU encoder **332** also encodes a flag (I_PCM_flag) which indicates whether or not the mode of encoding is the I_PCM mode and which is supplied from the PCM encoder **321**, and supplies the encoded flag as CU data to the accumulation buffer **308**. Furthermore, the CU encoder **332** encodes information about filter processing, such as an adaptive filter flag and a filter coefficient, supplied from the loop filter **312**, and supplies the encoded information as CU data to the accumulation buffer **308**.

(164) The method for encoding used by the CU encoder **332** is not specified (for example, CABAC, CAVLC, or the like). The NAL data and CU data supplied to the accumulation buffer **308** are combined together and accumulated therein.

(165) Note that the PCM encoder **321** actually controls whether or not the I_PCM mode is to be selected, by using the amount of code of encoded data which is generated by encoding, with the CU encoder **332**, a quantized orthogonal transform coefficient.

(166) Thus, for example, in a case where the I_PCM mode is not selected, the encoded data which has been supplied to the accumulation buffer **308** is adopted as an encoding result of the quantized orthogonal transform coefficient of the target CU. Thus, it is only necessary for the CU encoder **332** to encode additional information, such as an I_PCM_flag.

(167) In contrast, for example, in a case where the I_PCM mode is selected, the CU encoder **332** supplies the input pixel values of the target CU which are supplied from the screen rearrangement buffer **302** to the accumulation buffer **308**, as an encoding result (CU data). Thus, in this case, the encoded data of the target CU which has been supplied (encoded data generated by encoding the quantized orthogonal transform coefficient) is discarded. That is, all the processing operations regarding generation of the encoded data are redundant.

(168) As illustrated in FIG. **8**, the PCM encoder **321** includes an I_PCM_flag generator **341** and a PCM deciding unit **342**.

(169) The I_PCM_flag generator **341** generates an I_PCM_flag in accordance with a decision made by the PCM deciding unit **342**, and decides the value thereof. The I_PCM_flag generator **341** supplies the generated I_PCM_flag to the CU encoder **332** of the lossless encoder **307**. For example, in a case where the PCM deciding unit **342** selects the I_PCM mode, the I_PCM_flag generator **341** sets the value of the I_PCM_flag to be a value indicating that the I_PCM mode is selected (for example, “1”), and supplies the I_PCM_flag to the CU encoder **332**. Also, for example, for example, in a case where the PCM deciding unit **342** does not select the I_PCM mode, the I_PCM_flag generator **341** sets the value of the I_PCM_flag to be a value indicating that the I_PCM mode is not selected (for example, “0”), and supplies the I_PCM_flag to the CU encoder **332**.

(170) The PCM deciding unit **342** decides whether or not the mode of encoding is to be the I_PCM mode. The PCM deciding unit **342** obtains the amount of data of the input pixel values supplied from the screen rearrangement buffer **302**, compares the amount of data with the amount of generated code supplied from the rate controller **320**, and decides whether or not the I_PCM mode is to be selected on the basis of the comparison result. The PCM deciding unit **342** supplies an On/Off control signal representing the selection result to the CU encoder **332** and the I_PCM_flag generator **341**, thereby controlling an operation in accordance with the selection result.

(171) For example, in a case where the amount of data of the input pixel values is larger than the amount of generated code, the PCM deciding unit **342** does not select the I_PCM mode. In this case, the PCM deciding unit **342** supplies a control signal representing “On” to the CU encoder **332**, and causes the CU encoder **332** to encode the quantized orthogonal transform coefficient. Also, the PCM deciding unit **342** supplies a control signal representing “On” to the I_PCM_flag generator **341**, and causes the I_PCM_flag generator **341** to generate an I_PCM_flag having a value indicating that the I_PCM mode is not selected (for example, “0”).

(172) In contrast, for example, in a case where the amount of data of the input pixel values is smaller than or equal to the amount of generated code, the PCM deciding unit **342** selects the I_PCM mode. In this case, the PCM deciding unit **342** supplies a control signal representing “Off” to the CU encoder **332**, and causes the CU encoder **332** to output the input pixel values as an encoding result (CU data). Also, the PCM deciding unit **342** supplies a control signal representing “Off” to the I_PCM_flag generator **341**, and causes the I_PCM_flag generator **341** to generate an I_PCM_flag having a value indicating that the I_PCM mode is selected (for example “1”).

(173) The PCM deciding unit **342** is capable of deciding whether or not the I_PCM mode is to be

selected, in units of CUs of all the sizes (in an arbitrary layer) which are set in a sequence parameter set, as well as LCUs. Accordingly, for example, processing of suppressing the generation of many bits with a low QP by using the I_PCM mode can be executed in units of smaller CUs. Thus, the amount of code (including the data amount of non-encoded data in the I_PCM mode) can be controlled in more detail, and redundant processing which occurs in the I_PCM mode can be reduced.

(174) Also, the PCM deciding unit **342** supplies an On/Off control signal representing a selection result to the adaptive shift-to-left unit **303**, the adaptive shift-to-right unit **313**, and the adaptive shift-to-left unit **315**, thereby controlling IBDI in accordance with the selection result. That is, in a case where the mode of a target CU is the I_PCM mode, the PCM deciding unit **342** performs control so that processing of increasing and decreasing the bit precision is not performed by an adaptive shift device.

(175) For example, in a case where the I_PCM mode is not selected, the PCM deciding unit **342** supplies a control signal representing “On” to the adaptive shift-to-left unit **303**, the adaptive shift-to-right unit **313**, and the adaptive shift-to-left unit **315**, and causes shift-to-left processing and shift-to-right processing to be performed, so that the bit precision in internal processing is increased.

(176) In contrast, for example, in a case where the I_PCM mode is selected, the PCM deciding unit **342** supplies a control signal representing “Off” to the adaptive shift-to-left unit **303**, the adaptive shift-to-right unit **313**, and the adaptive shift-to-left unit **315**, and causes shift-to-left processing and shift-to-right processing to be skipped, so that the bit precision in internal processing is not increased.

(177) In the I_PCM mode, input image pixel values are transmitted to image compression information, and thus no arithmetic errors occur. Increasing the bit arithmetic precision therefor is redundant processing. The PCM deciding unit **342** is capable of eliminating such redundant processing by performing processing in the above-described manner.

(178) Furthermore, the PCM deciding unit **342** supplies the On/Off control signal representing the selection result to the loop filter **312**, thereby controlling adaptive loop filter processing (BALF) in accordance with the selection result. That is, in a case where the mode of a target CU is the I_PCM mode, the PCM deciding unit **342** performs control so that adaptive loop filter processing is not performed by the loop filter **312**.

(179) For example, in a case where the I_PCM mode is not selected, the PCM deciding unit **342** supplies a control signal representing “On” to the loop filter **312**, so that adaptive loop filter processing is performed. In contrast, for example, in a case where the I_PCM mode is selected, the PCM deciding unit **342** supplies a control signal representing “Off” to the loop filter **312**, so that adaptive loop filter processing is skipped.

(180) In the I_PCM mode, input image pixel values are transmitted to image compression information, and thus degradation does not occur. Performing adaptive loop filter processing thereon is redundant. The PCM deciding unit **342** is capable of eliminating such redundant processing by performing processing in the above-described manner.

(181) As illustrated in FIG. **8**, the loop filter **312** includes a deblocking filter **351**, a pixel sorting unit **352**, a filter coefficient calculator **353**, and a filtering unit **354**.

(182) As in the case of the deblocking filter **111**, the deblocking filter **351** performs deblocking filter processing on a decoded image (before-deblocking-filter pixel values) supplied from the computing unit **311**, thereby removing a block distortion.

(183) Even if a target CU to be processed is processed using the I_PCM mode, it is not always that the CU adjacent to the target CU has been processed using the I_PCM mode. Thus, even if the target CU is processed using the I_PCM mode, a block distortion may occur. Thus, deblocking filter processing is performed regardless of whether or not the mode of the target CU is the I_PCM mode.

(184) The deblocking filter **351** supplies the result of filter processing (after-deblocking-filter pixel values) to the pixel sorting unit **352**.

(185) The pixel sorting unit **352** sorts individual results of filter processing (after-deblocking-filter pixel values) as pixel values on which adaptive loop filter processing is to be performed or as pixel values on which adaptive loop filter processing is not to be performed, in accordance with the values of On/Off control signals supplied from the PCM deciding unit **342**.

(186) For example, in a case where a control signal representing “On” is supplied from the PCM deciding unit **342**, the pixel sorting unit **352** sorts the after-deblocking-filter pixel values of the corresponding CU as pixel values on which adaptive loop filter processing is to be performed. In contrast, for example, in a case where a control signal representing “Off” is supplied from the PCM deciding unit **342**, the pixel sorting unit **352** sorts the after-deblocking-filter pixel values of the corresponding CU as pixel values on which adaptive loop filter processing is not to be performed.

(187) The pixel sorting unit **352** supplies the sorted pixel values of individual pixels (after-deblocking-filter pixel values) to the filter coefficient calculator **353**.

(188) The filter coefficient calculator **353** calculates, using a Wiener filter, a filter coefficient (FIR filter coefficient) of an adaptive loop filter for the pixel values on which adaptive loop filter processing is to be performed among the after-deblocking-filter pixel values supplied thereto, so as to minimize the error with respect to an input image. That is, the filter coefficient calculator **353** calculates a filter coefficient by excluding pixels which are to be processed using the I_PCM mode.

(189) The filter coefficient calculator **353** supplies the after-deblocking-filter pixel values and the calculated filter coefficient to the filtering unit **354**.

(190) The filtering unit **354** performs, using the filter coefficient supplied thereto, adaptive loop filter processing on the pixel values which have been sorted as pixel values on which adaptive loop filter processing is to be performed. The filtering unit **354** supplies the pixel values on which filter processing has been performed and the pixel values which have been sorted as pixel values on which adaptive loop filter processing is not to be performed, which are after-adaptive-filter pixel values, to the adaptive shift-to-right unit **313**.

(191) Also, the filtering unit **354** generates an adaptive filter flag (on/off_flag), which is filter identification information indicating whether or not filter processing has been performed, for each of certain blocks which are set independently of CUs. The method for setting the value of the adaptive filter flag is not specified.

(192) For example, in a case where adaptive loop filter processing has been performed on some or all of the pixels in a target block to be processed (current block), the adaptive filter flag may be set to have a value indicating that filter processing has been performed (for example, “1”). Also, for example, in a case where adaptive loop filter processing has not been performed on all the pixels in the block, the adaptive filter flag may be set to have a value indicating that filter processing has not been performed (for example, “0”). The value of the adaptive filter flag may be set on the basis of another standard.

(193) The filtering unit **354** supplies the generated adaptive filter flag to the CU encoder **332** of the lossless encoder **307**, so that the adaptive filter flag is encoded and is provided to the decoding side. Note that, in a case where the value of the adaptive filter flag is a value indicating that filter processing has not been performed (for example, “0”), providing the adaptive filter flag to the decoding side may be skipped (the adaptive filter flag is not provided to the decoding side).

(194) For example, in a case where the value of the adaptive filter flag is a value indicating that filter processing has not been performed (for example, “0”) and where the coding format used by the lossless encoder **307** (CU encoder **332**) is VLC, the filtering unit **354** skips supply of the adaptive filter flag (does not provide the adaptive filter flag to the decoding side). Also, for example, in a case where the value of the adaptive filter flag is a value indicating that filter processing has not been performed (for example, “0”) and where the coding format used by the lossless encoder **307** (CU encoder **332**) is CABAC, the filtering unit **354** supplies the adaptive filter

flag to the CU encoder **332** of the lossless encoder **307** (provides the adaptive filter flag to the decoding side).

(195) This is because, in the case of VLC, if the amount of input information is small, it is possible to realize higher encoding efficiency, but in the case of CABAC, if the same information is continuously input, the probability at the time of performing arithmetic coding is biased, and higher encoding efficiency can be realized.

(196) Furthermore, the filtering unit **354** supplies the filter coefficient used for the adaptive loop filter processing to the CU encoder **332** of the lossless encoder **307**, so that the filter coefficient is encoded and provided to the decoding side. [PCM deciding unit]

(197) FIG. **9** is a block diagram illustrating a main example configuration of the PCM deciding unit **342** illustrated in FIG. **8**.

(198) As illustrated in FIG. **9**, the PCM deciding unit **342** includes an input data amount calculator **361**, a PCM determining unit **362**, an encoding controller **363**, an adaptive shift controller **364**, and a filter controller **365**.

(199) The input data amount calculator **361** calculates, for a target CU, the amount of input data which is the amount of data of input pixel values supplied from the screen rearrangement buffer **302**, and supplies the calculated amount of input data to the PCM determining unit **362**.

(200) The PCM determining unit **362** obtains the amount of generated code (generated bits) supplied from the rate controller **320**, compares the amount of generated code with the amount of input data supplied from the input data amount calculator **361**, and determines whether or not the I_PCM mode is to be selected for the CU on the basis of the comparison result. That is, the PCM determining unit **362** determines, for each CU in an arbitrary layer, whether or not the I_PCM mode is to be selected. The PCM determining unit **362** supplies the determination result to the encoding controller **363**, the adaptive shift controller **364**, and the filter controller **365**.

(201) On the basis of the determination result supplied from the PCM determining unit **362** (identification information indicating whether or not the I_PCM mode is selected), the encoding controller **363** supplies an On/Off control signal to the CU encoder **332** and the I_PCM_flag generator **341**.

(202) Thus, the encoding controller **363** is capable of controlling the mode of encoding in units of CUs in an arbitrary layer. Accordingly, the encoding controller **363** is capable of controlling the amount of code (including the data amount of non-encoded data in the I_PCM mode) in more detail, and is also capable of reducing redundant processing when the I_PCM mode is selected.

(203) On the basis of the determination result supplied from the PCM determining unit **362** (information indicating whether or not the I_PCM mode is selected), the adaptive shift controller **364** supplies an On/Off control signal to the adaptive shift-to-left unit **303**, the adaptive shift-to-right unit **313**, and the adaptive shift-to-left unit **315**.

(204) Thus, the adaptive shift controller **364** is capable of performing control so that the bit depth is not increased in internal arithmetic when the I_PCM mode is selected. Accordingly, the adaptive shift controller **364** is capable of reducing redundant processing.

(205) On the basis of the determination result supplied from the PCM determining unit **362** (information indicating whether or not the I_PCM mode is selected), the filter controller **365** supplies an On/Off control signal to the pixel sorting unit **352**.

(206) Thus, the filter controller **365** is capable of performing control so that adaptive loop filter processing is not performed when the I_PCM mode is selected. Accordingly, the filter controller **365** is capable of reducing redundant processing.

(207) As described above, the image encoding apparatus **300** is capable of reducing redundant processing and suppressing a decrease in the efficiency of encoding processing. Also, the image encoding apparatus **300** is capable of selecting the I_PCM mode (non-compression mode) in more detail (in units of smaller data units), and enhancing encoding efficiency. Therefore, the image encoding apparatus **300** is capable of enhancing encoding efficiency while suppressing a decrease

in the efficiency of encoding processing.

(208) [Flow of Encoding Processing]

(209) Next, description will be given of the flow of individual processing operations performed by the above-described image encoding apparatus **300**. First, an example of the flow of encoding processing will be described with reference to the flowchart in FIG. **10**.

(210) In step **S301**, the A/D converter **301** A/D-converts an input image. In step **S302**, the screen rearrangement buffer **302** stores the A/D-converted image therein and rearranges pictures from a display order to an encoding order.

(211) In step **S303**, the adaptive shift-to-left unit **303** adaptively performs shift to the left on the input image on the basis of control performed by the PCM encoder **321**. In step **S304**, the adaptive shift-to-left unit **315** adaptively performs shift to the left on a reference image.

(212) In step **S305**, the intra prediction unit **317** performs intra prediction processing in the intra prediction mode by using the reference image which is shifted to the left in step **S304**. In step **S306**, the motion prediction/compensation unit **318** performs inter motion prediction processing, in which motion prediction or motion compensation is preformed in the inter prediction mode, by using the reference image which is shifted to the left in step **S304**.

(213) Note that, actually, the processing of shifting the bit depth of the reference image to the left may be performed when the reference image is read out from the frame memory **314** in intra prediction processing or inter motion prediction processing.

(214) In step **S307**, the selector **319** determines an optimal mode on the basis of the individual cost functions which have been output from the intra prediction unit **317** and the motion prediction/compensation unit **318**. That is, the selector **319** selects any one of a prediction image generated by the intra prediction unit **317** and a prediction image generated by the motion prediction/compensation unit **318**.

(215) Also, selection information indicating which of the prediction images has been selected is supplied to, among the intra prediction unit **317** and the motion prediction/compensation unit **318**, the one corresponding to the selected prediction image. In a case where the prediction image of the optimal intra prediction mode is selected, the intra prediction unit **317** supplies intra prediction mode information indicating the optimal intra prediction mode and so forth to the lossless encoder **307**. In a case where the prediction image of the optimal inter prediction mode is selected, the motion prediction/compensation unit **318** outputs information indicating the optimal inter prediction mode, and if necessary, information based on the optimal inter prediction mode, to the lossless encoder **307**. Examples of the information based on the optimal inter prediction mode include motion vector information, flag information, and reference frame information.

(216) In step **S308**, the computing unit **304** computes the difference between the image in which the bit depth has been shifted to the left through the processing in step **S303** and the prediction image selected through the processing in step **S307**. The prediction image is supplied to the computing unit **304** via the selector **319** from the motion prediction/compensation unit **318** in a case where inter prediction is to be performed, and from the intra prediction unit **317** in a case where intra prediction is to be performed.

(217) The amount of difference data is smaller than that of the original image data. Thus, the amount of data can be reduced compared to a case where an image itself is encoded.

(218) In step **S309**, the orthogonal transform unit **305** performs an orthogonal transform on the difference information generated through the processing in step **S308**. Specifically, an orthogonal transform such as the discrete cosine transform or Karhunen-Loeve transform is performed, and a transform coefficient is output.

(219) In step **S310**, the quantizer **306** quantizes the orthogonal transform coefficient obtained through the processing in step **S309**.

(220) In step **S311**, the lossless encoder **307** encodes the transform coefficient quantized through the processing in step **S310**. That is, lossless coding such as variable-length coding or arithmetic

coding is performed on the difference image.

(221) The lossless encoder **307** encodes a quantization parameter calculated in step **S310** and adds the quantization parameter to encoded data. Also, the lossless encoder **307** encodes information about the mode of the prediction image selected through the processing in step **S307**, and adds the information to the encoded data which is obtained by encoding the difference image. That is, the lossless encoder **307** also encodes the optimal intra prediction mode information supplied from the intra prediction unit **317** or information based on the optimal inter prediction mode supplied from the motion prediction/compensation unit **318**, and adds the information to the encoded data.

(222) Furthermore, the lossless encoder **307** encodes the filter coefficient and flag information obtained from the loop filter **312** and adds them to the encoded data. Furthermore, the lossless encoder **307** encodes NAL data.

(223) In step **S312**, the accumulation buffer **308** accumulates the encoded data output from the lossless encoder **307**. The encoded data accumulated in the accumulation buffer **308** is read out as appropriate, and is transmitted to the decoding side via a transmission channel or a recording medium.

(224) In step **S313**, the rate controller **320** calculates the amount of code (the amount of generated code) of the encoded data accumulated in the accumulation buffer **308** through the processing in step **S312**, and controls the rate of the quantization operation performed by the quantizer **306** on the basis of the amount of code so that overflow or underflow does not occur. Also, the rate controller **320** supplies the amount of generated code to the PCM encoder **321**.

(225) In step **S314**, the PCM encoder **321** performs PCM encoding control processing by using the amount of generated code calculated in step **S313**. In step **S315**, the lossless encoder **307** performs PCM encoding processing in accordance with control performed by the PCM encoder **321**.

(226) In step **S316**, the dequantizer **309** to the frame memory **314** perform reference image generation processing in which the difference information quantized through the processing in step **S310** is locally decoded to generate a reference image.

(227) After the processing in step **S316** ends, the encoding processing ends. The encoding processing is repeatedly performed on, for example, individual CUs.

(228) [PCM Encoding Control Processing]

(229) Next, an example of the flow of the PCM encoding control processing which is performed in step **S314** in FIG. **10** will be described with reference to the flowchart in FIG. **11**.

(230) After the PCM encoding control processing starts, in step **S331**, the PCM determining unit **362** of the PCM deciding unit **342** obtains, from the rate controller **320**, the amount of generated code of the encoded data of the quantized orthogonal transform coefficient of the target CU.

(231) In step **S332**, the input data amount calculator **361** calculates the amount of input data of the input pixel values of the target CU.

(232) In step **S333**, the PCM determining unit **362** compares the amount of code obtained in step **S331** with the amount of input data calculated in step **S332**, and determines whether or not encoding is to be performed using the I_PCM mode.

(233) In step **S334**, the I_PCM_flag generator **341** generates an I_PCM_flag on the basis of an On/Off control signal which represents the determination result generated in step **S333** and which is supplied from the encoding controller **363**.

(234) In step **S335**, the encoding controller **363** supplies an On/Off control signal representing the determination result generated in step **S333** to the CU encoder **332**, thereby controlling encoding of CU data.

(235) In step **S336**, the adaptive shift controller **364** supplies an On/Off control signal representing the determination result generated in step **S333** to the adaptive shift-to-left unit **303**, the adaptive shift-to-right unit **313**, and the adaptive shift-to-left unit **315**, thereby controlling adaptive shift processing.

(236) In step **S337**, the encoding controller **363** supplies an On/Off control signal representing the

determination result generated in step S333 to the pixel sorting unit 352 of the loop filter 312, thereby controlling adaptive loop filter processing.

(237) After the processing in step S337 ends, the PCM deciding unit 342 ends the PCM encoding control processing, the processing returns to step S314 in FIG. 10, and the processing is performed from step S315.

(238) [Flow of PCM Encoding Processing]

(239) Next, an example of the flow of the PCM encoding processing which is performed in step S315 in FIG. 10 will be described with reference to the flowchart in FIG. 12.

(240) After the PCM encoding processing starts, in step S351, the CU encoder 332 determines whether or not encoding is to be performed using the I_PCM mode. In a case where control is performed in the above-described PCM encoding control processing so that encoding is to be performed using the I_PCM mode, the CU encoder 332 causes the processing to proceed to step S352. In step S352, the CU encoder 332 selects the input pixel values of the target CU as an encoding result. The CU encoder 332 causes the CU data of the target CU in the accumulation buffer 308 to be discarded, and causes the input pixel values to be accumulated in the accumulation buffer 308.

(241) After step S352 ends, the CU encoder 332 causes the processing to proceed to step S353. On the other hand, if it is determined in step S351 that encoding is not to be performed using the I_PCM mode, the CU encoder 332 causes the processing to proceed to step S353.

(242) In step S353, the CU encoder 332 encodes the I_PCM_flag which has been generated in the above-described PCM encoding control processing, and accumulates the I_PCM_flag in the accumulation buffer 308.

(243) After step S353 ends, the CU encoder 332 ends the PCM encoding processing, the processing returns to step S315 in FIG. 10, and the processing is performed from step S316. [Flow of reference image generation processing]

(244) Next, an example of the flow of the reference image generation processing which is performed in step S316 in FIG. 10 will be described with reference to the flowchart in FIG. 13.

(245) After the reference image generation processing starts, the adaptive shift-to-left unit 315 determines in step S371 whether or not the I_PCM mode is selected on the basis of control performed by the adaptive shift controller 364. If the I_PCM mode is selected, the bit depth is not increased in internal arithmetic, and thus processing is performed again starting from prediction processing. That is, if it is determined that the I_PCM mode is selected, the adaptive shift-to-left unit 315 causes the processing to proceed to step S372 without performing shift-to-left processing on a reference image.

(246) In step S372, the intra prediction unit 317 performs intra prediction processing using a reference image in which the bit depth is not shifted to the left. In step S373, the motion prediction/compensation unit 318 performs inter motion prediction processing using the reference image in which the bit depth is not shifted to the left.

(247) In step S374, the selector 319 decides an optimal mode on the basis of the individual cost function values output from the intra prediction unit 317 and the motion prediction/compensation unit 318. That is, the selector 319 selects any one of a prediction image generated by the intra prediction unit 317 and a prediction image generated by the motion prediction/compensation unit 318.

(248) Also, selection information indicating which of the prediction images has been selected is supplied to, among the intra prediction unit 317 and the motion prediction/compensation unit 318, the one corresponding to the selected prediction image. In a case where the prediction image of the optimal intra prediction mode is selected, the intra prediction unit 317 supplies intra prediction mode information indicating the optimal intra prediction mode and so forth to the lossless encoder 307. In a case where the prediction image of the optimal inter prediction mode is selected, the motion prediction/compensation unit 318 supplies information indicating the optimal inter

prediction mode, and if necessary, information based on the optimal inter prediction mode, to the lossless encoder **307**. Examples of the information based on the optimal inter prediction mode include motion vector information, flag information, and reference frame information.

(249) In a case where the I_PCM mode is selected, the bit depth in internal arithmetic is not increased. That is, shift-to-left processing performed on the reference image by the adaptive shift-to-left unit **303** is skipped. In step S375, the computing unit **304** computes the difference between the input image in which the bit depth is not shifted to the left and the prediction image selected through the processing in step S374. The prediction image is supplied to the computing unit **304** via the selector **319** from the motion prediction/compensation unit **318** in a case where inter prediction is performed and from the intra prediction unit **317** in a case where intra prediction is performed.

(250) In step S376, the orthogonal transform unit **305** performs an orthogonal transform on the difference information generated through the processing in step S375. Specifically, an orthogonal transform such as the discrete cosine transform or Karhunen-Loeve transform is performed, and a transform coefficient is output. In step S377, the quantizer **306** quantizes the orthogonal transform coefficient obtained through the processing in step S376.

(251) After the orthogonal transform coefficient has been quantized, the quantizer **306** causes the processing to proceed to step S378, where a reference image is generated using the orthogonal transform coefficient (data on which shift-to-left processing has not been performed) quantized in step S377.

(252) In contrast, if it is determined in step S371 that the I_PCM mode is not selected, the adaptive shift-to-left unit **315** skips the processing operations in step S372 to step S377, and causes the processing to proceed to step S378. That is, in a case where the I_PCM mode is not selected, a reference image is generated using the orthogonal transform coefficient (data on which shift-to-left processing has been performed) quantized in step S310 in FIG. 10.

(253) In step S378, the dequantizer **309** dequantizes the quantized orthogonal transform coefficient (also referred to as quantized coefficient) in accordance with the characteristic corresponding to the characteristic of the quantizer **306**. In step S379, the inverse orthogonal transform unit **310** performs inverse orthogonal transform on the orthogonal transform coefficient obtained through the processing in step S378, in accordance with the characteristic corresponding to the characteristic of the orthogonal transform unit **305**.

(254) In step S380, the computing unit **311** adds the prediction image to the locally decoded difference information, thereby generating a locally decoded image (an image corresponding to the input to the computing unit **304**). For example, in a case where the I_PCM mode is selected, the computing unit **311** adds a prediction image on which shift-to-left processing has not been performed to difference information on which shift-to-left processing has not been performed, so as to generate a decoded image on which shift-to-left processing has not been performed. Also, for example, in a case where the I_PCM mode is not selected, the computing unit **311** adds a prediction image on which shift-to-left processing has been performed to difference information on which shift-to-left processing has been performed, so as to generate a decoded image on which shift-to-left processing has been performed.

(255) In step S381, the loop filter **312** performs loop filter processing on the locally decoded image obtained through the processing in step S380 on the basis of control performed by the filter controller **365**, and performs loop filter processing including deblocking filter processing, adaptive loop filter processing, and so forth as appropriate.

(256) In step S382, the adaptive-shift-to-right unit **313** determines whether or not the I_PCM mode is selected, on the basis of control performed by the adaptive shift controller **364**. If it is determined that the I_PCM mode is not selected, the adaptive shift-to-right unit **313** causes the processing to proceed to step S383.

(257) In a case where the I_PCM mode is not selected, the decoded image has undergone an

increase in the bit depth in internal arithmetic. Thus, in step S383, the adaptive shift-to-right unit 313 adaptively performs shift to right on the bit depth of the result of filter processing (decoded image) obtained through the loop filter processing in step S381. After the processing in step S383 ends, the adaptive shift-to-right unit 313 causes the processing to proceed to step S384.

(258) If it is determined in step S382 that the I_PCM mode is selected, the adaptive shift-to-right unit 313 causes the processing to proceed to step S384 without performing shift-to-right processing.

(259) In step S384, the frame memory 314 stores the decoded image. After the processing in step S384 ends, the frame memory 314 ends the reference image generation processing, the processing returns to step S316 in FIG. 10, and the encoding processing ends.

(260) [Flow of Loop Filter Processing]

(261) Next, an example of the flow of the loop filter processing which is performed in step S381 in FIG. 13 will be described with reference to the flowchart in FIG. 14.

(262) After the loop filter processing starts, in step S401, the deblocking filter 351 of the loop filter 312 performs deblocking filter processing on the decoded image (before-deblocking-filter pixel values) supplied from the computing unit 311.

(263) In step S402, the pixel sorting unit 352 sorts the individual pixels of the decoded image on the basis of whether or not the mode is the I_PCM mode, under control performed by the filter controller 365 of the PCM deciding unit 342.

(264) In step S403, the filter coefficient calculator 353 calculates a filter coefficient for the pixels which have been sorted to undergo filter processing (target pixels to be processed). In step S404, the filtering unit 354 performs adaptive filter processing on the target pixels to be processed, using the filter coefficient calculated in step S403.

(265) In step S405, the filtering unit 354 sets an adaptive filter flag for the target block to be processed, and supplies the adaptive filter flag and the filter coefficient to the CU encoder 332, so that the adaptive filter flag and the filter coefficient are encoded.

(266) After the processing in step S405 ends, the loop filter 312 ends the loop filter processing, the processing returns to step S381 in FIG. 13, and the processing is performed from step S382.

(267) By performing the individual processing operations in the above-described manner, the image encoding apparatus 300 is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

2. Second Embodiment

(268) [Image Decoding Apparatus]

(269) FIG. 15 is a block diagram illustrating a main example configuration of an image decoding apparatus. The image decoding apparatus 500 illustrated in FIG. 15 is an apparatus which is basically similar to the image decoding apparatus 200 illustrated in FIG. 2, and decodes encoded data which is generated by encoding image data.

(270) The image decoding apparatus 500 illustrated in FIG. 15 is a decoding apparatus corresponding to the image encoding apparatus 300 illustrated in FIG. 7. Encoded data which has been encoded by the image encoding apparatus 300 is supplied to the image decoding apparatus 500 via an arbitrary path, for example, a transmission channel, a recording medium, or the like, and is decoded.

(271) As illustrated in FIG. 15, the image decoding apparatus 500 includes an accumulation buffer 501, a lossless decoder 502, a dequantizer 503, an inverse orthogonal transform unit 504, a computing unit 505, a loop filter 506, an adaptive shift-to-right unit 507, a screen rearrangement buffer 508, and a D/A converter 509. Also, the image decoding apparatus 500 includes a frame memory 510, an adaptive shift-to-left unit 511, a selector 512, an intra prediction unit 513, a motion prediction/compensation unit 514, and a selector 515.

(272) The image decoding apparatus 500 further includes a PCM decoder 516.

(273) The accumulation buffer 501 accumulates encoded data transmitted thereto, as in the case of

the accumulation buffer **201**. The encoded data has been encoded by the image encoding apparatus **300**.

(274) The lossless decoder **502** reads out encoded data from the accumulation buffer **501** at a certain timing, and decodes the encoded data using a format corresponding to the coding format used by the lossless encoder **307** illustrated in FIG. **15**. At this time, the lossless decoder **502** supplies the I_PCM_flag included in the encoded data to the PCM decoder **516**, and causes the PCM decoder **516** to determine whether or not the mode is the I_PCM mode (non-compression mode).

(275) In a case where the mode is the I_PCM mode, the CU data obtained from the accumulation buffer **501** is non-encoded data. Thus, the lossless decoder **502** supplies the CU data to the dequantizer **503** in accordance with control performed by the PCM decoder **516**.

(276) In a case where the mode is not the I_PCM mode, the CU data obtained from the accumulation buffer **501** is encoded data. Thus, the lossless decoder **502** decodes the CU data in accordance with control performed by the PCM decoder **516**, and supplies a decoding result to the dequantizer **503**.

(277) Note that, for example, in a case where the target CU has been intra-coded, intra prediction mode information is stored in the header portion of the encoded data. The lossless decoder **502** also decodes the intra prediction mode information, and supplies the information to the intra prediction unit **513**. On the other hand, in a case where the target CU has been inter-coded, motion vector information and inter prediction mode information are stored in the header portion of the encoded data. The lossless decoder **502** also decodes the motion vector information and the inter prediction mode information, and supplies the information to the motion prediction/compensation unit **514**.

(278) In a case where the mode is not the I_PCM mode, the dequantizer **503** dequantizes the coefficient data (quantized coefficient) supplied from the lossless decoder **502** using a method corresponding to the quantization method used by the quantizer **306** illustrated in FIG. **7**, as in the case of the dequantizer **203**. That is, the dequantizer **503** dequantizes the quantized coefficient using a method similar to that used by the dequantizer **309** illustrated in FIG. **7**. The dequantizer **503** supplies the dequantized coefficient data, that is, the orthogonal transform coefficient, to the inverse orthogonal transform unit **504**.

(279) Also, in a case where the mode is the I_PCM mode, the dequantizer **503** supplies the CU data supplied from the lossless decoder **502** (image data which is not encoded) to the inverse orthogonal transform unit **504**.

(280) In a case where the mode is not the I_PCM mode, the inverse orthogonal transform unit **504** performs inverse orthogonal transform on the orthogonal transform coefficient using a method corresponding to the orthogonal transform method used by the orthogonal transform unit **305** illustrated in FIG. **7** (a method similar to that used by the inverse orthogonal transform unit **310** illustrated in FIG. **7**), as in the case of the inverse orthogonal transform unit **204**. With the inverse orthogonal transform processing, the inverse orthogonal transform unit **504** obtains decoded residual data corresponding to the residual data before orthogonal transform is performed in the image encoding apparatus **300**. For example, fourth-order inverse orthogonal transform is performed. The inverse orthogonal transform unit **504** supplies the decoded residual data obtained through inverse orthogonal transform to the computing unit **505**.

(281) Also, in a case where the mode is the I_PCM mode, the inverse orthogonal transform unit **504** supplies the CU data supplied from the dequantizer **503** (image data which is not encoded) to the computing unit **505**.

(282) Also, a prediction image is supplied to the computing unit **505** from the intra prediction unit **513** or the motion prediction/compensation unit **514** via the selector **515**.

(283) In a case where the mode is not the I_PCM mode, the computing unit **505** adds the decoded residual data and the prediction image, and thereby obtains decoded image data corresponding to the image data before the prediction image is subtracted by the computing unit **304** of the image

encoding apparatus **300**, as in the case of the computing unit **205**. The computing unit **505** supplies the decoded image data to the loop filter **506**.

(284) In a case where the mode is the I_PCM mode, the computing unit **505** supplies the CU data supplied from the inverse orthogonal transform unit **504** (image data which is not encoded) to the loop filter **506**. In this case, the CU data is not residual information, and thus addition to the prediction image is not necessary.

(285) The loop filter **506** performs loop filter processing including deblocking filter processing, adaptive loop filter processing, or the like, on the decoded image supplied from the computing unit **505** as appropriate, under control performed by the PCM decoder **516**.

(286) More specifically, the loop filter **506** performs deblocking filter processing similar to that performed by the deblocking filter **206** on the decoded image, thereby removing a block distortion of the decoded image. Also, the loop filter **506** performs loop filter processing using a Wiener filter on the result of the deblocking filter processing (decoded image from which a block distortion has been removed) in accordance with control performed by the PCM decoder **516**, thereby improving image quality.

(287) Alternatively, the loop filter **506** may perform arbitrary filter processing on the decoded image. Alternatively, the loop filter **506** may perform filter processing by using the filter coefficient supplied from the image encoding apparatus **300** illustrated in FIG. 7.

(288) The loop filter **506** supplies the result of filter processing (the decoded image on which filter processing has been performed) to the adaptive shift-to-right unit **507**.

(289) The adaptive shift-to-right unit **507** is a processing unit similar to the adaptive shift-to-right unit **313** (FIG. 7), and is controlled by the PCM decoder **516**. In a case where the mode is not the I_PCM mode, the adaptive shift-to-right unit **507** shifts the decoded image data supplied from the loop filter **506** in the right direction, so as to reduce the bit depth by a certain number of bits (for example, 4 bits), for example, the bit depth is decreased from 12 bits to 8 bits. That is, the adaptive shift-to-right unit **507** shifts the decoded image data to the right by the number of bits by which the image data has been shifted to the left in the image encoding apparatus **300**, so as to change the bit depth of the image data to the state where shift to the left has not been performed (the state at the time when the image data is read out from the screen rearrangement buffer **302** (FIG. 7)).

(290) The adaptive shift-to-right unit **507** supplies the image data on which shift-to-right processing has been performed to the screen rearrangement buffer **508** and the frame memory **510**.

(291) Note that the amount of shift to the right (the amount of bits) is not specified as long as the amount is the same as the amount of shift to the left in the image encoding apparatus **300** (adaptive shift-to-left unit **303**). That is, the amount may be fixed or variable. For example, the amount of shift may be predetermined (shared in advance between the image encoding apparatus **300** and the image decoding apparatus **500**), or the image decoding apparatus **500** may be allowed to calculate the amount of shift in the image encoding apparatus **300**, or information indicating the amount of shift may be provided from the image encoding apparatus **300**.

(292) Alternatively, the shift-to-right processing may be skipped in accordance with control performed by the PCM decoder **516**. For example, in the case of the I_PCM mode, image data does not undergo shift-to-left processing in the image encoding apparatus **300**. Therefore, in this case, the adaptive shift-to-right unit **507** is controlled by the PCM decoder **516**, and supplies the decoded image data supplied from the loop filter **506** to the screen rearrangement buffer **508** and the frame memory **510** without shifting the decoded image data in the right direction.

(293) The screen rearrangement buffer **508** rearranges images, as in the case of the screen rearrangement buffer **207**. That is, the frames which have been rearranged in an encoding order by the screen rearrangement buffer **302** illustrated in FIG. 7 are rearranged in the original display order. The screen rearrangement buffer **508** supplies the decoded image data of individual frames to the D/A converter **509** in the original display order.

(294) The D/A converter **509** D/A-converts the frame images supplied from the screen

rearrangement buffer **508**, and outputs the frame images to a display (not illustrated) so that the frame images are displayed, as in the case of the D/A convertor **208**.

(295) The frame memory **510** stores the decoded image supplied from the adaptive shift-to-right unit **507**, and supplies the stored decoded image as a reference image to the adaptive shift-to-left unit **511** at a certain timing.

(296) The adaptive shift-to-left unit **511** is a processing unit similar to the adaptive shift-to-left unit **315**, is controlled by the PCM decoder **516**, shifts the image data (reference image) read out from the frame memory **510** in the left direction as appropriate, and increases the bit depth thereof by a certain number of bits (for example, 4 bits).

(297) For example, in a case where the mode is not the I_PCM mode, the decoded image data supplied from the inverse orthogonal transform unit **504** to the computing unit **505** is image data which has been shifted to the left in the image encoding apparatus **300** (for example, the bit depth is 12 bits). Thus, the adaptive shift-to-left unit **511** increases the bit depth of the reference image which has been read out from the frame memory **510** by a certain number of bits (for example, increases the bit depth from 8 bits to 12 bits) in accordance with control performed by the PCM decoder **516**.

(298) Then, the adaptive shift-to-left unit **511** supplies the image data on which shift-to-left processing has been performed to the selector **512**. As a result of increasing the bit depth in this way, the bit depth of the reference image can be made the same as the bit depth of the decoded image, so that the reference image can be added to the decoded image. Also, the accuracy of internal arithmetic such as prediction processing can be increased, and errors can be suppressed.

(299) On the other hand, for example, in a case where the mode is the I_PCM mode, the decoded image data supplied from the inverse orthogonal transform unit **504** to the computing unit **505** is image data on which shift-to-left processing has not been performed in the image encoding apparatus **300** (for example, the bit depth is 8 bits). Thus, the adaptive shift-to-left unit **511** supplies the reference image read out from the frame memory **510** to the selector **512** without increasing the bit depth in accordance with control performed by the PCM decoder **516**.

(300) In the case of intra prediction, the selector **512** supplies the reference image supplied from the adaptive shift-to-left unit **511** to the intra prediction unit **513**. On the other hand, in the case of inter prediction, the selector **512** supplies the reference image supplied from the adaptive shift-to-left unit **511** to the motion prediction/compensation unit **514**.

(301) The intra prediction unit **513** is supplied with information indicating the intra prediction mode, which has been obtained by decoding header information, and so forth from the lossless decoder **502** as appropriate. The intra prediction unit **513** performs intra prediction using the reference image obtained from the frame memory **510** using the intra prediction mode used by the intra prediction unit **317**, and generates a prediction image. That is, the intra prediction unit **513** is also capable of performing intra prediction using an arbitrary mode other than the modes defined in the AVC coding format, like the intra prediction unit **317**.

(302) The intra prediction unit **513** supplies the generated prediction image to the selector **515**.

(303) The motion prediction/compensation unit **514** is supplied with information obtained by decoding header information (prediction mode information, motion vector information, reference frame information, flag, various parameters, etc.) from the lossless decoder **502**.

(304) The motion prediction/compensation unit **514** performs inter prediction using the reference image obtained from the frame memory **510** using the inter prediction mode used by the motion prediction/compensation unit **318**, thereby generating a prediction image. That is, the motion prediction/compensation unit **514** is also capable of performing intra prediction using an arbitrary mode other than the modes defined in the AVC coding format, like the motion prediction/compensation unit **318**.

(305) The motion prediction/compensation unit **514** supplies the generated prediction image to the selector **515**.

(306) The selector **515** selects the prediction image generated by the motion prediction/compensation unit **514** or the intra prediction unit **513**, and supplies the prediction image to the computing unit **505**, as in the case of the selector **213**.

(307) The PCM decoder **516** controls the lossless decoder **502**, the loop filter **506**, the adaptive shift-to-right unit **507**, and the adaptive shift-to-left unit **511** on the basis of the I_PCM_flag supplied from the lossless decoder **502**.

(308) [Lossless Decoder, PCM Decoder, and Loop Filter]

(309) FIG. **16** is a block diagram illustrating a main example configuration of the lossless decoder **502**, the PCM decoder **516**, and the loop filter **506** illustrated in FIG. **15**.

(310) As illustrated in FIG. **16**, the lossless decoder **502** includes a NAL decoder **531** and a CU decoder **532**. The NAL decoder **531** decodes encoded NAL data supplied from the accumulation buffer **501**, and supplies the decoded NAL data to the CU decoder **532**.

(311) The CU decoder **532** decodes encoded CU data which is supplied from the accumulation buffer **501**, on the basis of the NAL data supplied from the NAL decoder **531**.

(312) Also, upon obtaining an I_PCM_flag by decoding the CU data, the CU decoder **532** supplies the I_PCM_flag to the PCM decoder **516**. The CU decoder **532** decodes the encoded CU data in accordance with control performed by the PCM decoder **516** based on the I_PCM_flag (on the basis of an On/Off control signal supplied from the PCM encoder **321**).

(313) For example, in a case where the I_PCM_flag has a value indicating that the mode is not the I_PCM mode, and where a control signal representing “On” is obtained from the PCM decoder **516**, the CU decoder **532** decodes the encoded CU data supplied from the accumulation buffer **501** on the basis of the NAL data supplied from the NAL decoder **531**, and supplies a quantized orthogonal transform coefficient to the dequantizer **503**.

(314) On the other hand, in a case where the I_PCM_flag has a value indicating that the mode is the I_PCM mode, and where a control signal representing “Off” is obtained from the PCM decoder **516**, the CU decoder **532** supplies the CU data supplied from the accumulation buffer **501** (image data which is not encoded) to the dequantizer **503**.

(315) Note that the CU decoder **532** supplies information obtained by decoding the CU data, such as a filter coefficient and an adaptive filter flag, to the loop filter **506**.

(316) As illustrated in FIG. **8**, the PCM decoder **516** includes an I_PCM_flag buffer **541** and a PCM controller **542**.

(317) The I_PCM_flag buffer **541** stores the I_PCM_flag supplied from the CU decoder **532** of the lossless decoder **502**, and supplies the I_PCM_flag to the PCM controller **542** at a certain timing.

(318) The PCM controller **542** determines whether or not the I_PCM mode was selected at the time of encoding in the image encoding apparatus **300**, on the basis of the value of the I_PCM_flag obtained from the I_PCM_flag buffer **541**. On the basis of the determination result, the PCM controller **542** supplies an On/Off control signal to the CU decoder **532**, and controls the operation thereof.

(319) For example, in a case where it is determined that the I_PCM mode was not selected, the PCM controller **542** supplies a control signal representing “On” to the CU decoder **532**, causes the CU decoder **532** to decode the encoded CU data, and causes the CU decoder **532** to supply a quantized orthogonal transform coefficient obtained through the decoding to the dequantizer **503**.

(320) On the other hand, for example, in a case where it is determined that the I_PCM mode was selected, the PCM controller **542** supplies a control signal representing “Off” to the CU decoder **532**, and causes the CU decoder **532** to supply CU data, which is non-encoded data, to the dequantizer **503** as output pixel values.

(321) With the above-described control, the CU decoder **532** is capable of appropriately decoding encoded data supplied from the image encoding apparatus **300**. That is, the CU decoder **532** is capable of decoding encoded data more appropriately even if whether or not the mode is the I_PCM mode is controlled in units of CUs, which are smaller than LCUs.

(322) Also, the PCM controller **542** supplies an On/Off control signal to the adaptive shift-to-right unit **507** and the adaptive shift-to-left unit **511** on the basis of the determination result about the I_PCM mode, and controls the operation thereof.

(323) For example, in a case where it is determined that the I_PCM mode was not selected, the PCM controller **542** supplies a control signal representing “On” to the adaptive shift-to-right unit **507** and the adaptive shift-to-left unit **511**, and causes shift-to-left processing and shift-to-right processing to be performed, so that the bit precision in internal processing is increased.

(324) On the other hand, in a case where it is determined that the I_PCM mode was selected, the PCM controller **542** supplies a control signal representing “Off” to the adaptive shift-to-right unit **507** and the adaptive shift-to-left unit **511**, and causes shift-to-left processing and shift-to-right processing to be skipped, so that the bit precision in internal processing is not increased.

(325) With the above-described control, the PCM controller **542** is capable of appropriately eliminating redundant processing even if whether or not the mode is the I_PCM mode is controlled in units of CUs, which are smaller than LCUs.

(326) Furthermore, the PCM controller **542** supplies an On/Off control signal to the loop filter **506** on the basis of the determination result about the I_PCM mode, and controls the operation thereof. For example, in a case where it is determined that the I_PCM mode was not selected, the PCM controller **542** supplies a control signal representing “On” to the loop filter **506**. On the other hand, for example, in a case where it is determined that the I_PCM mode was selected, the PCM controller **542** supplies a control signal representing “Off” to the loop filter **506**.

(327) With the above-described control, the PCM controller **542** is capable of eliminating redundant processing more appropriately even if whether or not the mode is the I_PCM mode is controlled in units of CUs, which are smaller than LCUs.

(328) As illustrated in FIG. **8**, the loop filter **506** includes a deblocking filter **551**, a pixel sorting unit **552**, and a filtering unit **553**.

(329) The deblocking filter **551** performs deblocking filter processing on the decoded image supplied from the computing unit **505** (before-deblocking-filter pixel values), thereby removing a block distortion, as in the case of the deblocking filter **206**.

(330) As in the case of the image encoding apparatus **300**, deblocking filter processing is performed regardless of whether or not the mode for the target CU is the I_PCM mode. The deblocking filter **551** supplies the result of filter processing (after-deblocking-filter pixel values) to the pixel sorting unit **552**.

(331) In accordance with the value of the On/Off control signal supplied from the PCM controller **542**, the pixel sorting unit **552** sorts individual results of filter processing (after-deblocking-filter pixel values) as pixel values on which adaptive loop filter processing is to be performed and pixel values on which adaptive loop filter processing is not to be performed.

(332) For example, in a case where a control signal representing “On” is supplied from the PCM controller **542**, the pixel sorting unit **552** sorts the after-deblocking-filter pixel values of the CU as pixel values on which adaptive loop filter processing is to be performed. In contrast, for example, in a case where a control signal representing “Off” is supplied from the PCM controller **542**, the pixel sorting unit **552** sorts the after-deblocking-filter pixel values of the CU as pixel values on which adaptive loop filter processing is not to be performed.

(333) The pixel sorting unit **552** supplies the sorted pixel values of individual pixels (after-deblocking-filter pixel values) to the filtering unit **553**.

(334) The filtering unit **553** performs adaptive loop filter processing on the pixel values which have been sorted as pixel values on which adaptive loop filter processing is to be performed, on the basis of the adaptive loop filter flag supplied from the CU decoder **532**.

(335) Adaptive loop filter processing is performed in units of certain blocks which are set independently of CUs. In a case where the value of the adaptive loop filter flag of the target block is a value indicating that filter processing has not been performed on the encoding side (for

example, “0”), the filtering unit **553** skips adaptive loop filter processing for the target block, and supplies the after-deblocking-filter pixel values supplied thereto to the adaptive shift-to-right unit **507** as after-adaptive-filter pixel values.

(336) By the way, the method for setting an adaptive loop filter flag depends on the specification and so forth of the image encoding apparatus **300**. Thus, depending on the method for setting an adaptive loop filter flag by the image encoding apparatus **300**, even if the value of the adaptive loop filter flag of the target block is a value indicating that filter processing has been performed on the encoding side (for example, “1”), there is a possibility that a pixel of the I_PCM mode is included in the target block.

(337) Thus, in a case where there is a pixel value which has been sorted as a pixel value on which adaptive loop filter processing is not to be performed in the target block, the filtering unit **553** skips adaptive loop filter processing for the target block even if the value of the adaptive loop filter flag of the target block is a value indicating that filter processing has been performed on the encoding side (for example, “1”), and supplies the after-deblocking-filter pixel values supplied thereto to the adaptive shift-to-right unit **507** as after-adaptive-filter pixel values.

(338) That is, the filtering unit **553** performs adaptive loop filter processing on the target block only in a case where the value of the adaptive loop filter flag of the target block is a value indicating that filter processing has been performed on the encoding side (for example, “1”) and where all the pixels in the target block have been sorted as pixels on which adaptive loop filter processing is to be performed.

(339) Note that, in the case of performing adaptive loop filter processing, the filtering unit **553** performs adaptive loop filter processing using the filter coefficient supplied from the CU decoder **32** (the filter coefficient used in the adaptive loop filter processing on the encoding side).

(340) The filtering unit **553** supplies the pixel values on which adaptive loop filter processing has been performed to the adaptive shift-to-right unit **507** as after-adaptive-filter pixel values.

(341) With the above-described control, the PCM controller **542** is capable of appropriately eliminating redundant processing even if whether or not the mode is the I_PCM mode (non-compression mode) is controlled in units of CUs, which are smaller than LCUs.

(342) As described above, with the processing operations performed by the individual processing units, the image decoding apparatus **500** is capable of realizing enhanced encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(343) [Flow of Decoding Processing]

(344) Next, the flow of individual processing operations performed by the above-described image decoding apparatus **500** will be described. First, an example of the flow of decoding processing will be described with reference to the flowcharts in FIG. **17** and FIG. **18**.

(345) After decoding processing starts, in step **S501**, the accumulation buffer **501** accumulates an image (encoded data) transmitted thereto. In step **S502**, the CU decoder **532** obtains an adaptive filter flag from the encoded data accumulated in step **S501**. In step **S503**, the CU decoder **532** obtains an I_PCM_flag from the encoded data accumulated in step **S501**. The I_PCM_flag buffer **541** stores the I_PCM_flag.

(346) In step **S504**, the PCM controller **542** determines whether or not the encoding mode of the encoded data (CU data) accumulated in step **S501** is the I_PCM mode (that is, non-encoded data), on the basis of the value of the I_PCM_flag stored in the I_PCM_flag buffer **541**.

(347) If it is determined that the encoding mode is not the I_PCM mode, the PCM controller **542** causes the processing to proceed to step **S505**. In step **S505**, the lossless decoder **502** performs lossless decoding processing, decodes the encoded data (CU data) accumulated in step **S501**, and obtains a quantized orthogonal transform coefficient, filter coefficient, and so forth.

(348) In step **S506**, the dequantizer **503** dequantizes the quantized orthogonal transform coefficient, which is obtained through the processing in step **S505**. In step **S507**, the inverse orthogonal transform unit **504** performs inverse orthogonal transform on the orthogonal transform coefficient

which is dequantized through the processing in step S506, and generates decoded image data.

(349) In step S508, the adaptive shift-to-left unit 511 obtains the reference image corresponding to the target CU from the frame memory 510, and performs shift-to-left processing on the reference image in accordance with control performed by the PCM decoder 516.

(350) In step S509, the intra prediction unit 513 or the motion prediction/compensation unit 514 performs prediction processing using the reference image on which shift-to-left processing is performed in step S508, thereby generating a prediction image.

(351) In step S510, the computing unit 505 adds the prediction image generated by the intra prediction unit 513 or the motion prediction/compensation unit 514 in step S509, to the difference information obtained through the processing in step S507.

(352) In step S511, the loop filter 506 performs loop filter processing on the addition result obtained in step S510.

(353) In step S512, the adaptive shift-to-right unit 507 performs shift-to-right processing on the result of loop filter processing obtained in step S511, in accordance with control performed by the PCM decoder 516.

(354) In step S513, the frame memory 510 stores the decoded image data as a reference image.

(355) In step S514, the screen rearrangement buffer 508 rearranges the frames of the decoded image data. That is, the frames of the decoded image data which have been rearranged in an encoding order by the screen rearrangement buffer 302 (FIG. 7) of the image encoding apparatus 300 are rearranged in the original display order.

(356) In step S515, the D/A converter 509 D/A-converts the decoded image data in which the frames are rearranged in step S514. The decoded image data is output to a display (not illustrated) and the image thereof is displayed.

(357) Also, if it is determined in step S504 that the encoding mode is the I_PCM mode, the PCM controller 542 causes the processing to proceed to step S521 in FIG. 18. In step S521, the lossless decoder 502 performs lossless decoding processing, and regards the encoded data (non-compressed data) accumulated in step S501 as an encoding result (output pixel values).

(358) In step S522, the loop filter 506 performs loop filter processing on the output pixel values obtained in step S521. After the processing in step S522 ends, the loop filter 506 returns the processing to step S513 in FIG. 17, so that the subsequent step is performed.

(359) [Flow of Loop Filter Processing]

(360) Next, an example of the flow of the loop filter processing which is performed in step S511 in FIG. 17 and step S522 in FIG. 18 will be described with reference to the flowchart in FIG. 19.

(361) After the loop filter processing starts, in step S541, the deblocking filter 551 performs deblocking filter processing on the before-deblocking-filter pixel values which are obtained in step S510 or step S521.

(362) In step S542, the filtering unit 553 obtains a filter coefficient. In step S543, the pixel sorting unit 552 determines whether or not adaptive loop filtering is to be performed, on the basis of the value of the adaptive loop filter flag. In a case where it is determined that adaptive loop filter processing is to be performed, the pixel sorting unit 552 causes the processing to proceed to step S544.

(363) In step S544, the pixel sorting unit 552 sorts the after-deblocking-filter pixel values in accordance with whether or not the mode is the I_PCM mode, in accordance with control performed by the PCM controller 542.

(364) In step S545, the filtering unit 545 performs adaptive loop filter processing on the after-deblocking-filter pixel values which have been sorted to undergo adaptive loop filter processing, by using the filter coefficient obtained in step S542. After the processing in step S545 ends, the filtering unit 545 ends the loop filter processing, the processing returns to step S511 in FIG. 17 or step S522 in FIG. 18, and the processing proceeds to step S512 in FIG. 17 or step S513 in FIG. 17.

(365) If it is determined in step S543 in FIG. 19 that adaptive loop filter processing is not to be

performed, the pixel sorting unit **552** ends the loop filter processing, the processing returns to step **S511** in FIG. **17** or step **S522** in FIG. **18**, and the processing proceeds to step **S512** in FIG. **17** or step **S513** in FIG. **17**.

(366) As a result of performing the individual processing operations in the above-described manner, the image decoding apparatus **500** is capable of realizing enhanced encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(367) [Position of I_PCM Information]

(368) As described above, selection control of the I_PCM mode can be performed in units of CUs, so that encoding using the I_PCM mode can be performed on only some of CUs in an LCU, as illustrated in part A of FIG. **20**, for example. Part A of FIG. **20** illustrates the structure of CUs in one LCU. The LCU illustrated in part A of FIG. **20** is constituted by seven CUs, that is, CU**0** to CU**6**. The individual numerals indicate the processing order in the LCU. In the example illustrated in part A of FIG. **20**, the shaded CUs (CU**1**, CU**3**, and CU**6**) are to be encoded using the I_PCM mode.

(369) In this case, as illustrated in part B of FIG. **20**, I_PCM information, which is information regarding I_PCM inside the LCU, may be added at the top of the data of LCU in a code stream.

(370) The I_PCM information includes, for example, flag information indicating whether or not the LCU includes a CU which is to be encoded using the I_PCM mode. In the case of the example illustrated in FIG. **20**, CU**1**, CU**3**, and CU**6** are CUs to be encoded using the I_PCM mode, and thus the flag information is set to be a value indicating that the LCU includes a CU which is to be encoded using the I_PCM mode (for example, “1”).

(371) In this way, the image encoding apparatus **300** adds information about I_PCM inside the LCU at the top of the LCU, and thereby the image decoding apparatus **500** is capable of easily determining whether or not the LCU includes a CU which is to be encoded using the I_PCM mode, before decoding the LCU.

(372) Also, the I_PCM information includes, for example, information indicating a CU which is to be encoded using the I_PCM mode included in the LCU. In the case of the example illustrated in FIG. **20**, CU**1**, CU**3**, and CU**6** are CUs which are to be encoded using the I_PCM mode, and thus the I_PCM information includes information indicating these CUs (CU**1**, CU**3**, and CU**6**).

(373) In this way, the image encoding apparatus **300** adds information about I_PCM inside the LCU at the top of the LCU, and thereby the image decoding apparatus **500** is capable of easily determining which CU in the LCU has been encoded using the I_PCM mode (non-compression mode), before decoding the LCU.

(374) Note that, in the description given above, pieces of information other than image data, such as an I_PCM_flag, an adaptive filter flag, a filter coefficient, and I_PCM information, are provided from the image encoding apparatus **300** to the image decoding apparatus **500** as necessary, but these pieces of information may be added at an arbitrary position of encoded data. For example, the pieces of information may be added at the top of a CU or LCU, or may be added to a slice header, or may be stored in a sequence parameter set (SPS), a picture parameter set (PPS), or the like.

Alternatively, for example, the pieces of information may be stored in a parameter set (for example, a header of a sequence or picture) of an SEI (Supplemental Enhancement Information) or the like.

(375) Furthermore, the pieces of information may be transmitted to the decoding side separately from encoded data. In that case, it is necessary to clarify (allow the decoding side to determine) the correspondence between the pieces of information and encoded data, but the method therefor is not specified. For example, table information indicating the correspondence may be separately created, or link information indicating the other side may be embedded in the data of each side.

(376) Alternatively, the pieces of information may be shared in advance by the image encoding apparatus **300** and the image decoding apparatus **500**. In that case, transmission of the pieces of information may be omitted.

3. Third Embodiment

(377) [Personal Computer]

(378) The above-described series of processing operations may be executed by hardware or may be executed by software. In this case, for example, the hardware or software may be constituted as the personal computer illustrated in FIG. 21.

(379) In FIG. 21, a CPU (Central Processing Unit) **601** of a personal computer **600** executes various processing operations in accordance with a program stored in a ROM (Read Only Memory) **602** or a program loaded from a storage unit **613** to a RAM (Random Access Memory) **603**. Also, data which is necessary for the CPU **601** to execute various processing operations is stored in the RAM **603** as appropriate.

(380) The CPU **601**, the ROM **602**, and the RAM **603** are connected to one another via a bus **604**. Also, an input/output interface **610** is connected to the bus **604**.

(381) An input unit **611** including a keyboard, a mouse, or the like; an output unit **612** including a display such as a CRT (Cathode Ray Tube) or an LCD (Liquid Crystal Display), and a speaker or the like; the storage unit **613** including a hard disk or the like; and a communication unit **614** including a modem or the like are connected to the input/output interface **610**. The communication unit **614** performs communication processing via a network, including the Internet.

(382) Also, a drive **615** is connected to the input/output interface **610** as necessary, a removable medium **621** such as a magnetic disk, an optical disc, a magneto-optical disc, or a semiconductor memory is loaded thereto as appropriate, and a computer program read out therefrom is installed in the storage unit **613** as necessary.

(383) In the case of causing software to execute the above-described series of processing operations, a program constituting the software is installed via a network or a recording medium.

(384) The recording medium is constituted by, for example, as illustrated in FIG. 21, the removable medium **621** which is provided separately from the main body of the apparatus to distribute a program to a user, which contains the program recorded thereon, and which is constituted by a magnetic disk (including a flexible disk), an optical disc (including a CD-ROM (Compact Disc-Read Only Memory) and a DVD (Digital Versatile Disc)), a magneto-optical disc (including an MD (Mini Disc)), or a semiconductor memory. Alternatively, the recording medium is constituted by the ROM **602** containing the program recorded thereon or a hard disk included in the storage unit **613**, which is provided to a user in the state of being incorporated in the main body of the apparatus in advance.

(385) The program executed by the computer may be a program in which processing operations are performed in time series in accordance with the order described in this description, or may be a program in which processing operations are performed in parallel or at necessary timings, for example, when the processing operations are called.

(386) Also, in this description, the steps describing a program recorded on a recording medium may be processing operations which are performed in time series in accordance with the described order, or may be processing operations which are executed in parallel or individually.

(387) Also, in this description, a “system” is an entire apparatus constituted by a plurality of devices.

(388) Also, the configuration described above as a single device (or processing unit) may be divided into a plurality of devices (or processing units). In contrast, the configuration described above as a plurality of devices (or processing units) may be combined into a single device (or processing unit). Also, a configuration other than that described above may of course be added to the configuration of the individual devices (or processing units). Furthermore, as long as the configuration and operation of the entire system are substantially the same, part of the configuration of a certain device (or processing unit) may be included in the configuration of another device (or another processing unit). That is, the embodiments of the present technology are not limited to the above-described embodiments, and various modifications can be made without deviating from the gist of the present technology.

(389) For example, each of the lossless encoder **307**, the loop filter **312**, and the PCM encoder **321** illustrated in FIG. **8** may be configured as an independent device. Also, each of the NAL encoder **331**, the CU encoder **332**, the I_PCM_flag generator **341**, the PCM deciding unit **342**, the deblocking filter **351**, the pixel sorting unit **352**, the filter coefficient calculator **353**, and the filtering unit **354** illustrated in FIG. **8** may be configured as an independent device.

(390) Furthermore, each of the input data amount calculator **361**, the PCM determining unit **362**, the encoding controller **363**, the adaptive shift controller **364**, and the filter controller illustrated in FIG. **9** may be configured as an independent device.

(391) Alternatively, these processing units may be arbitrarily combined together to constitute an independent device. Of course, these processing units may be combined with an arbitrary processing unit illustrated in FIG. **7** to FIG. **9**, or may be combined with a processing unit which is not illustrated.

(392) This is the same in the image decoding apparatus **500**. For example, each of the lossless decoder **502**, the loop filter **506**, and the PCM decoder **516** illustrated in FIG. **15** may be configured as an independent device. Also, each of the NAL decoder **531**, the CU decoder **532**, the I_PCM_flag buffer **541**, the PCM controller **542**, the deblocking filter **551**, the pixel sorting unit **552**, and the filtering unit **553** illustrated in FIG. **16** may be configured as an independent device.

(393) Furthermore, these processing units may be arbitrarily combined together to constitute an independent device. Of course, these processing units may be combined with an arbitrary processing unit illustrated in FIG. **15** and FIG. **16**, or may be combined with a processing unit which is not illustrated.

(394) Also, for example, the above-described image encoding apparatus and image decoding apparatus can be applied to an arbitrary electronic apparatus. Hereinafter, examples thereof will be described.

4. Fourth Embodiment

(395) [Television Receiver]

(396) FIG. **22** is a block diagram illustrating a main example configuration of a television receiver that includes the image decoding apparatus **500**.

(397) The television receiver **1000** illustrated in FIG. **22** includes a terrestrial tuner **1013**, a video decoder **1015**, a video signal processing circuit **1018**, a graphic generation circuit **1019**, a panel drive circuit **1020**, and a display panel **1021**.

(398) The terrestrial tuner **1013** receives a broadcast wave signal of terrestrial analog broadcasting via an antenna, decodes the signal, obtains a video signal, and supplies the video signal to be video decoder **1015**. The video decoder **1015** performs decoding processing on the video signal supplied from the terrestrial tuner **1013**, and supplies a digital component signal obtained thereby to the video signal processing circuit **1018**.

(399) The video signal processing circuit **1018** performs certain processing, such as noise reduction, on the video data supplied from the video decoder **1015**, and supplies the obtained video data to the graphic generation circuit **1019**.

(400) The graphic generation circuit **1019** generates, for example, video data of a program which is to be displayed on the display panel **1021**, or generates image data by performing processing based on an application supplied via a network, and supplies the generated video data or image data to the panel drive circuit **1020**. Also, the graphic generation circuit **1019** performs, as appropriate, processing of generating video data (graphic) for displaying a screen which is used by a user to select an item or the like, and supplying video data which is obtained by superposing the generated video data on video data of a program to the panel drive circuit **1020**.

(401) The panel drive circuit **1020** drives the display panel **1021** on the basis of the data supplied from the graphic generation circuit **1019**, and causes the display panel **1021** to display video of a program or the above-described various screens.

(402) The display panel **1021** is constituted by an LCD (Liquid Crystal Display) or the like, and

displays video of a program and so forth in accordance with control performed by the panel drive circuit **1020**.

(403) Also, the television receiver **1000** includes an audio A/D (Analog/Digital) converter circuit **1014**, an audio signal processing circuit **1022**, an echo cancellation/audio synthesis circuit **1023**, an audio amplifier circuit **1024**, and a speaker **1025**.

(404) The terrestrial tuner **1013** demodulates a received broadcast wave signal, and thereby obtains an audio signal as well as a video signal. The terrestrial tuner **1013** supplies the obtained audio signal to the audio A/D converter circuit **1014**.

(405) The audio A/D converter circuit **1014** performs A/D conversion processing on the audio signal supplied from the terrestrial tuner **1013**, and supplies a digital audio signal obtained thereby to the audio signal processing circuit **1022**.

(406) The audio signal processing circuit **1022** performs certain processing, such as noise reduction, on the audio data supplied from the audio A/D converter circuit **1014**, and supplies audio data obtained thereby to the echo cancellation/audio synthesis circuit **1023**.

(407) The echo cancellation/audio synthesis circuit **1023** supplies the audio data supplied from the audio signal processing circuit **1022** to the audio amplifier circuit **1024**.

(408) The audio amplifier circuit **1024** performs D/A conversion processing and amplification processing on the audio data supplied from the echo cancellation/audio synthesis circuit **1023**, adjusts the audio data so as to have a certain volume, and causes audio to be output from the speaker **1025**.

(409) Furthermore, the television receiver **1000** includes a digital tuner **1016** and an MPEG decoder **1017**.

(410) The digital tuner **1016** receives a broadcast wave signal of digital broadcasting (terrestrial digital broadcasting, BS (Broadcasting Satellite)/CS (Communications Satellite) digital broadcasting) via an antenna, demodulates the signal, obtains an MPEG-TS (Moving Picture Experts Group-Transport Stream), and supplies it to the MPEG decoder **1017**.

(411) The MPEG decoder **1017** descrambles the MPEG-TS supplied from the digital tuner **1016**, and extracts a stream including the data of a program as a target to be reproduced (target to be viewed and listened to). The MPEG decoder **1017** decodes the audio packets constituting the extracted stream and supplies the audio data obtained thereby to the audio signal processing circuit **1022**, and also decodes the video packets constituting the stream and supplies video data obtained thereby to the video signal processing circuit **1018**. Also, the MPEG decoder **1017** supplies EPG (Electronic Program Guide) data extracted from the MPEG-TS to a CPU **1032** via a path which is not illustrated.

(412) The television receiver **1000** includes the above-described image decoding apparatus **500** serving as the MPEG decoder **1017** that decodes video packets in this way. Note that the MPEG-TS transmitted from a broadcast station or the like has been encoded by the image encoding apparatus **300**.

(413) As in the case of the image decoding apparatus **500**, the MPEG decoder **1017** appropriately decodes encoded data for which the selection of the I_PCM mode is controlled in units of CUs, which are smaller than LCUs. Thus, the MPEG decoder **1017** is capable of realizing a decrease in redundant processing for encoding and a decrease in redundant information included in the encoded data. Accordingly, the MPEG decoder **1017** is capable of realizing enhanced encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(414) As in the case of the video data supplied from the video decoder **1015**, the video data supplied from the MPEG decoder **1017** undergoes certain processing in the video signal processing circuit **1018**, video data or the like generated in the graphic generation circuit **1019** is superposed thereon as appropriate, the video data is supplied to the display panel **1021** via the panel drive circuit **1020**, and the image thereof is displayed.

(415) As in the case of the audio data supplied from the audio A/D converter circuit **1014**, the audio

data supplied from the MPEG decoder **1017** undergoes certain processing in the audio signal processing circuit **1022**, is supplied to the audio amplifier circuit **1024** via the echo cancellation/audio synthesis circuit **1023**, and undergoes D/A conversion processing and amplification processing. As a result, audio adjusted to have a certain volume is output from the speaker **1025**.

(416) Also, the television receiver **1000** includes a microphone **1026** and an A/D converter circuit **1027**.

(417) The A/D converter circuit **1027** receives a signal of user's voice captured by the microphone **1026** that is provided in the television receiver **1000** for voice conversation, performs A/D conversion processing on the received audio signal, and supplies obtained digital audio data to the echo cancellation/audio synthesis circuit **1023**.

(418) In a case where audio data of a user (user A) of the television receiver **1000** is supplied from the A/D converter circuit **1027**, the echo cancellation/audio synthesis circuit **1023** performs echo cancellation on the audio data of the user A, and causes audio data which is obtained through synthesis with other audio data to be output from the speaker **1025** via the audio amplifier circuit **1024**.

(419) Furthermore, the television receiver **1000** includes an audio codec **1028**, an internal bus **1029**, an SDRAM (Synchronous Dynamic Random Access Memory) **1030**, a flash memory **1031**, the CPU **1032**, a USB (Universal Serial Bus) I/F **1033**, and a network I/F **1034**.

(420) The A/D converter circuit **1027** receives a signal of user's voice captured by the microphone **1026** that is provided in the television receiver **1000** for voice conversation, performs A/D conversion processing on the received audio signal, and supplies obtained digital audio data to the audio codec **1028**.

(421) The audio codec **1028** converts the audio data supplied from the A/D converter circuit **1027** into data of a certain format for transmitting it via a network, and supplies the audio data to the network I/F **1034** via the internal bus **1029**.

(422) The network I/F **1034** is connected to a network via a cable attached to a network terminal **1035**. The network I/F **1034** transmits audio data supplied from the audio codec **1028** to another apparatus connected to the network, for example. Also, the network I/F **1034** receives, via the network terminal **1035**, audio data transmitted from another apparatus connected via the network, for example, and supplies the audio data to the audio codec **1028** via the internal bus **1029**.

(423) The audio codec **1028** converts the audio data supplied from the network I/F **1034** into data of a certain format, and supplies the data to the echo cancellation/audio synthesis circuit **1023**.

(424) The echo cancellation/audio synthesis circuit **1023** performs echo cancellation on the audio data supplied from the audio codec **1028**, and causes audio data obtained through synthesis with other audio data to be output from the speaker **1025** via the audio amplifier circuit **1024**.

(425) The SDRAM **1030** stores various pieces of data necessary for the CPU **1032** to perform processing.

(426) The flash memory **1031** stores a program executed by the CPU **1032**. The program stored in the flash memory **1031** is read out by the CPU **1032** at a certain timing, for example, at the start-up of the television receiver **1000**. The flash memory **1031** also stores EPG data obtained via digital broadcasting and data obtained from a certain server via a network.

(427) For example, the flash memory **1031** stores an MPEG-TS including content data obtained from a certain server via a network under control performed by the CPU **1032**. The flash memory **1031** supplies the MPEG-TS to the MPEG decoder **1017** via the internal bus **1029**, for example, under control performed by the CPU **1032**.

(428) The MPEG decoder **1017** processes the MPEG-TS, as in the case of the MPEG-TS supplied from the digital tuner **1016**. In this way, the television receiver **1000** is capable of receiving content data of video, audio, or the like via a network, decoding the data using the MPEG decoder **1017**, and causing the video to be displayed or the audio to be output.

(429) Also, the television receiver **1000** includes a light receiver **1037** that receives an infrared signal transmitted from a remote control **1051**.

(430) The light receiver **1037** receives an infrared ray from the remote control **1051**, and outputs a control code representing the detail of a user operation obtained through demodulation to the CPU **1032**.

(431) The CPU **1032** executes a program stored in the flash memory **1031**, and controls the entire operation of the television receiver **1000** in accordance with a control code or the like supplied from the light receiver **1037**. The CPU **1032** is connected to the individual units of the television receiver **1000** via paths which are not illustrated.

(432) The USB I/F **1033** transmits data to/receives data from an external apparatus of the television receiver **1000**, the apparatus being connected via a USB cable attached to a USB terminal **1036**.

The network I/F **1034** connects to a network via a cable attached to the network terminal **1035**, and transmits/receives data other than audio data to/from various apparatuses connected to the network.

(433) The television receiver **1000** includes the image decoding apparatus **500** serving as the MPEG decoder **1017**, thereby being capable of realizing enhancement of encoding efficiency of content data while suppressing a decrease in the efficiency of encoding processing at the time of generating the content data, which is obtained via a broadcast wave signal received via an antenna or a network.

5. Fifth Embodiment

(434) [Mobile Phone]

(435) FIG. **23** is a block diagram illustrating a main example configuration of a mobile phone that includes the image encoding apparatus **300** and the image decoding apparatus **500**.

(436) The mobile phone **1100** illustrated in FIG. **23** includes a main controller **1150** configured to collectively control individual units, a power supply circuit unit **1151**, an operation input controller **1152**, an image encoder **1153**, a camera I/F unit **1154**, an LCD controller **1155**, an image decoder **1156**, a multiplexer/demultiplexer unit **1157**, a recording/reproducing unit **1162**, a modulation/demodulation circuit unit **1158**, and an audio codec **1159**. These are connected to one another via a bus **1160**.

(437) Also, the mobile phone **1100** includes an operation key **1119**, a CCD (Charge Coupled Devices) camera **1116**, a liquid crystal display **1118**, a storage unit **1123**, a transmission/reception circuit unit **1163**, an antenna **1114**, a microphone **1121**, and a speaker **1117**.

(438) When a call ends or a power key is turned on through a user operation, the power supply circuit unit **1151** supplies power from a battery pack to the individual units, thereby bringing the mobile phone **1100** into an operable state.

(439) The mobile phone **1100** performs various operations, such as transmission/reception of an audio signal, transmission/reception of an electronic mail or image data, image capturing, or data recording, in various modes, such as an audio call mode or a data communication mode, on the basis of control performed by the main controller **1150**, which includes a CPU, a ROM, a RAM, and so forth.

(440) For example, in the audio call mode, the mobile phone **1100** converts an audio signal collected by the microphone **1121** into digital audio data using the audio codec **1159**, performs spectrum spread processing thereon using the modulation/demodulation circuit unit **1158**, and performs digital-to-analog conversion processing and frequency conversion processing using the transmission/reception circuit unit **1163**. The mobile phone **1100** transmits a signal to be transmitted obtained through the conversion processing to a base station (not illustrated) via the antenna **1114**. The signal to be transmitted (audio signal) transmitted to the base station is supplied to a mobile phone of the other end of a call via a public telephone line network.

(441) Also, for example, in the audio call mode, the mobile phone **1100** amplifies a reception signal received by the antenna **1114** using the transmission/reception circuit unit **1163**, further performs frequency conversion processing and analog-to-digital conversion processing, performs spectrum

inverse spread processing using the modulation/demodulation circuit unit **1158**, and converts the signal into an analog audio signal using the audio codec **1159**. The mobile phone **1100** outputs the analog audio signal obtained through the conversion from the speaker **1117**.

(442) Furthermore, for example, in the case of transmitting an electronic mail in the data communication mode, the mobile phone **1100** accepts, in the operation input controller **1152**, the text data of the electronic mail input through an operation of the operation key **1119**. The mobile phone **1100** processes the text data using the main controller **1150**, and causes the text data to be displayed as an image on the liquid crystal display **1118** via the LCD controller **1155**.

(443) Also, the mobile phone **1100** generates, in the main controller **1150**, electronic mail data on the basis of the text data or a user instruction accepted by the operation input controller **1152**. The mobile phone **1100** performs spectrum spread processing on the electronic mail data using the modulation/demodulation circuit unit **1158**, and performs digital-to-analog conversion processing and frequency conversion processing using the transmission/reception circuit unit **1163**. The mobile phone **1100** transmits a signal to be transmitted obtained through the conversion processing to a base station (not illustrated) via the antenna **1114**. The signal to be transmitted (electronic mail) transmitted to the base station is supplied to a certain destination via a network and a mail server or the like.

(444) Also, for example, in the case of receiving an electronic mail in the data communication mode, the mobile phone **1100** receives a signal transmitted from the base station via the antenna **1114** using the transmission/reception circuit unit **1163**, amplifies the signal, and further performs frequency conversion processing and analog-to-digital conversion processing. The mobile phone **1100** performs spectrum inverse spread processing on the received signal using the modulation/demodulation circuit unit **1158** to restore original electronic mail data. The mobile phone **1100** displays the restored electronic mail data on the liquid crystal display **1118** via the LCD controller **1155**.

(445) Additionally, the mobile phone **1100** is also capable of causing the received electronic mail data to be recorded (stored) in the storage unit **1123** via the recording/reproducing unit **1162**.

(446) The storage unit **1123** is an arbitrary rewritable storage medium. The storage unit **1123** may be, for example, a semiconductor memory, such as a RAM or a built-in flash memory, a hard disk, or a removable medium, such as a magnetic disk, a magneto-optical disc, an optical disc, a USB memory, or a memory card. Of course, other types of media may be used.

(447) Furthermore, for example, in the case of transmitting image data in the data communication mode, the mobile phone **1100** generates image data through image capturing using the CCD camera **1116**. The CCD camera **1116** includes optical devices, such as a lens and a diaphragm, and a CCD serving as a photoelectric conversion element, captures an image of a subject, converts the intensity of received light into an electric signal, and generates image data of the image of the subject. The CCD camera **1116** encodes the image data using the image encoder **1153** via the camera I/F unit **1154**, thereby converting the image data into encoded image data.

(448) The mobile phone **1100** includes the above-described image encoding apparatus **300** serving as the image encoder **1153** that performs the above-described processing. As in the case of the image encoding apparatus **300**, the image encoder **1153** controls the selection of the I_PCM mode in units of CUs, which are smaller than LCUs. That is, the image encoder **1153** is capable of further reducing redundant processing for encoding, and is also capable of further reducing redundant information included in encoded data. Accordingly, the image encoder **1153** is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(449) In addition, at the same time, the mobile phone **1100** performs, in the audio codec **1159**, analog-to-digital conversion on audio collected by the microphone **1121** during image capturing by the CCD camera **1116**, and also encodes it.

(450) The mobile phone **1100** multiplexes, in the multiplexer/demultiplexer unit **1157**, the encoded

image data supplied from the image encoder **1153** and the digital audio data supplied from the audio codec **1159** using a certain method. The mobile phone **1100** performs spectrum spread processing on the multiplexed data obtained as a result using the modulation/demodulation circuit unit **1158**, and performs digital-to-analog conversion processing and frequency conversion processing using the transmission/reception circuit unit **1163**. The mobile phone **1100** transmits a signal to be transmitted obtained through the conversion processing to a base station (not illustrated) via the antenna **1114**. The signal to be transmitted (image data) which has been transmitted to the base station is supplied to the other end of communication via a network or the like.

(451) Note that, in the case of not transmitting image data, the mobile phone **1100** is capable of causing the image data generated by the CCD camera **1116** to be displayed on the liquid crystal display **1118** via the LCD controller **1155**, not via the image encoder **1153**.

(452) Also, for example, in the case of receiving data of a moving image file that is linked to a simple web page or the like in the data communication mode, the mobile phone **1100** receives a signal transmitted from a base station via the antenna **1114** using the transmission/reception circuit unit **1163**, amplifies the signal, and further performs frequency conversion processing and analog-to-digital conversion processing thereon. The mobile phone **1100** performs spectrum inverse spread processing on the received signal to restore original multiplexed data using the modulation/demodulation circuit unit **1158**. The mobile phone **1100** demultiplexes the multiplexed data into encoded image data and audio data using the multiplexer/demultiplexer unit **1157**.

(453) The mobile phone **1100** decodes the encoded image data using the image decoder **1156** to generate reproduced moving image data, and causes the data to be displayed on the liquid crystal display **1118** via the LCD controller **1155**. Accordingly, for example, the moving image data included in the moving image file linked to the simple web page is displayed on the liquid crystal display **1118**.

(454) The mobile phone **1100** includes the above-described image decoding apparatus **500** serving as the image decoder **1156** for performing such processing. That is, as in the case of the image decoding apparatus **500**, the image decoder **1156** appropriately decodes encoded data for which the selection of the I_PCM mode is controlled in units of CUs, which are smaller than LCUs. Thus, the image decoder **1156** is capable of realizing a decrease in redundant processing for encoding and a decrease in redundant information included in the encoded data. Accordingly, the image decoder **1156** is capable of realizing enhanced encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(455) At this time, the mobile phone **1100** converts digital audio data into an analog audio signal using the audio codec **1159**, and causes the signal to be output from the speaker **1117**. Accordingly, for example, audio data included the moving image file linked to the simple web page is reproduced.

(456) Note that, as in the case of an electronic mail, the mobile phone **1100** is also capable of causing the received data linked to the simple web page or the like to be recorded (stored) in the storage unit **1123** via the recording/reproducing unit **1162**.

(457) Also, the mobile phone **1100** is capable of analyzing a two-dimensional code obtained by the CCD camera **1116** through image capturing and obtaining information recorded in the two-dimensional code using the main controller **1150**.

(458) Furthermore, the mobile phone **1100** is capable of communicating with an external apparatus through an infrared ray using an infrared communication unit **1181**.

(459) By including the image encoding device **300** serving as the image encoder **1153**, the mobile phone **1100** is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing, for example, when encoding and transmitting image data generated by the CCD camera **1116**.

(460) Also, by including the image decoding apparatus **500** serving as the image decoder **1156**, the

mobile phone **1100** is capable of realizing enhanced encoding efficiency of data while suppressing a decrease in the efficiency of encoding processing at the time of generating the data (encoded data) of a moving image file linked to a simple web page or the like.

(461) Note that, although a description has been given above that the mobile phone **1100** includes the CCD camera **1116**, an image sensor using a CMOS (Complementary Metal Oxide Semiconductor) (CMOS image sensor) may be used instead of the CCD camera **1116**. In this case, too, the mobile phone **1100** is capable of capturing an image of a subject and generating image data of the image of the subject, as in the case of using the CCD camera **1116**.

(462) Also, although a description has been given above of the mobile phone **1100**, the image encoding apparatus **300** and the image decoding apparatus **500** can be applied to any apparatus having an image capturing function and a communication function similar to those of the mobile phone **1100**, such as a PDA (Personal Digital Assistants), a smart phone, a UMPC (Ultra Mobile Personal Computer), a net book, or a notebook personal computer, as in the case of the mobile phone **1100**.

6. Sixth Embodiment

(463) [Hard Disk Recorder]

(464) FIG. **24** is a block diagram illustrating a main example configuration of a hard disk recorder that includes the image encoding apparatus **300** and the image decoding apparatus **500**.

(465) The hard disk recorder (HDD recorder) **1200** illustrated in FIG. **24** is an apparatus that stores, in a hard disk built therein, audio data and video data of a broadcast program included in a broadcast wave signal (television signal) that is transmitted by a satellite, an antenna on the ground, or the like and that is received by a tuner, and that provides the stored data to a user at a timing corresponding to an instruction provided by the user.

(466) The hard disk recorder **1200** is capable of, for example, extracting audio data and video data from a broadcast wave signal, appropriately decoding them, and causing them to be stored in the hard disk built therein. Also, the hard disk recorder **1200** is capable of, for example, obtaining audio data and video data from another apparatus via a network, appropriately decoding them, and causing them to be stored in the hard disk built therein.

(467) Furthermore, the hard disk recorder **1200** is capable of, for example, decoding audio data and video data recorded on the hard disk built therein, supplying them to a monitor **1260**, causing the image thereof to be displayed on the screen of the monitor **1260**, and causing the audio thereof to be output from the speaker of the monitor **1260**. Also, the hard disk recorder **1200** is capable of, for example, decoding audio data and video data extracted from a broadcast wave signal obtained via a tuner or audio data and video data obtained from another apparatus via a network, supplying them to the monitor **1260**, causing the image thereof to be displayed on the screen of the monitor **1260**, and causing the audio thereof to be output from the speaker of the monitor **1260**.

(468) Of course, another operation can be performed.

(469) As illustrated in FIG. **24**, the hard disk recorder **1200** includes a receiving unit **1221**, a demodulating unit **1222**, a demultiplexer **1223**, an audio decoder **1224**, a video decoder **1225**, and a recorder controller **1226**. The hard disk recorder **1200** further includes an EPG data memory **1227**, a program memory **1228**, a working memory **1229**, a display converter **1230**, an OSD (On Screen Display) controller **1231**, a display controller **1232**, a recording/reproducing unit **1233**, a D/A converter **1234**, and a communication unit **1235**.

(470) Also, the display converter **1230** includes a video encoder **1241**. The recording/reproducing unit **1233** includes an encoder **1251** and a decoder **1252**.

(471) The receiving unit **1221** receives an infrared signal from a remote control (not illustrated), converts the signal into an electric signal, and outputs the electric signal to the recorder controller **1226**. The recorder controller **1226** is constituted by, for example, a microprocessor or the like, and executes various processing operations in accordance with a program stored in the program memory **1228**. At this time, the recorder controller **1226** uses the working memory **1229** as

necessary.

(472) The communication unit **1235** is connected to a network, and performs communication processing with another apparatus via the network. For example, the communication unit **1235** is controlled by the recorder controller **1226**, communicates with a tuner (not illustrated), and outputs a channel selection control signal mainly to the tuner.

(473) The demodulating unit **1222** demodulates the signal supplied from the tuner and outputs the signal to the demultiplexer **1223**. The demultiplexer **1223** demultiplexes the data supplied from the demodulating unit **1222** into audio data, video data, and EPG data, and outputs them to the audio decoder **1224**, the video decoder **1225**, and the recorder controller **1226**, respectively.

(474) The audio decoder **1224** decodes the audio data input thereto, and outputs the audio data to the recording/reproducing unit **1233**. The video decoder **1225** decodes the video data input thereto, and outputs the video data to the display converter **1230**. The recorder controller **1226** supplies the EPG data input thereto to the EPG data memory **1227** so as to store it therein.

(475) The display converter **1230** encodes, with the video encoder **1241**, the video data supplied from the video decoder **1225** or the recorder controller **1226** into video data of an NTSC (National Television Standards Committee) format, for example, and outputs the video data to the recording/reproducing unit **1233**. Also, the display converter **1230** converts the size of the screen of the video data supplied from the video decoder **1225** or the recorder controller **1226** into the size corresponding to the size of the monitor **1260**, converts the video data into video data of the NTSC format with the video encoder **1241**, converts the video data into an analog signal, and outputs the analog signal to the display controller **1232**.

(476) The display controller **1232** superposes the OSD signal output from the OSD (On Screen Display) controller **1231** on the video signal input from the display converter **1230** under control performed by the recorder controller **1226**, outputs it to the display of the monitor **1260**, and causes it to be displayed thereon.

(477) Also, the monitor **1260** is supplied with the audio data which has been output from the audio decoder **1224** and which has been converted into an analog signal by the D/A converter **1234**. The monitor **1260** outputs this audio signal from the speaker built therein.

(478) The recording/reproducing unit **1233** includes a hard disk serving as a storage medium for holding video data, audio data, and the like recorded thereon.

(479) The recording/reproducing unit **1233** encodes, with the encoder **1251**, the audio data supplied from the audio decoder **1224**, for example. Also, the recording/reproducing unit **1233** encodes, with the encoder **1251**, the video data supplied from the video encoder **1241** of the display converter **1230**. The recording/reproducing unit **1233** combines, with a multiplexer, the encoded data of the audio data and the encoded data of the video data. The recording/reproducing unit **1233** performs channel coding on the composite data to amplify it, and writes the data on the hard disk via a recording head.

(480) The recording/reproducing unit **1233** reproduces the data recorded on the hard disk via a reproducing head, amplifies the data, and demultiplexes the data into audio data and video data using a demultiplexer. The recording/reproducing unit **1233** decodes, with the decoder **1252**, the audio data and the video data. The recording/reproducing unit **1233** D/A-converts the decoded audio data and outputs the audio data to the speaker of the monitor **1260**. Also, the recording/reproducing unit **1233** D/A-converts the decoded video data and outputs the video data to the display of the monitor **1260**.

(481) The recorder controller **1226** reads out the latest EPG data from the EPG data memory **1227** on the basis of a user instruction represented by an infrared signal which is supplied from the remote control and which is received via the receiving unit **1221**, and supplies the EPG data to the OSD controller **1231**. The OSD controller **1231** generates image data corresponding to the input EPG data, and outputs the image data to the display controller **1232**. The display controller **1232** outputs the video data input from the OSD controller **1231** to the display of the monitor **1260**, and

causes the video data to be displayed thereon. Accordingly, an EPG (electronic program guide) is displayed on the display of the monitor **1260**.

(482) Also, the hard disk recorder **1200** is capable of obtaining various pieces of data, such as video data, audio data, or EPG data, supplied from another apparatus via a network, such as the Internet.

(483) The communication unit **1235** is controlled by the recorder controller **1226**, obtains encoded data of video data, audio data, EPG data, and so forth transmitted from another apparatus via a network, and supplies the encoded data to the recorder controller **1226**. For example, the recorder controller **1226** supplies the obtained encoded data of video data and audio data to the recording/reproducing unit **1233**, and causes the hard disk to store the encoded data. At this time, the recorder controller **1226** and the recording/reproducing unit **1233** may perform processing, such as re-encoding, as necessary.

(484) Also, the recorder controller **1226** decodes the obtained encoded data of video data and audio data, and supplies the obtained video data to the display converter **1230**. The display converter **1230** processes the video data supplied from the recorder controller **1226**, like the video data supplied from the video decoder **1225**, supplies the video data to the monitor **1260** via the display controller **1232**, and causes the image thereof to be displayed thereon.

(485) Also, in accordance with display of the image, the recorder controller **1226** may supply decoded audio data to the monitor **1260** via the D/A converter **1234** and cause the audio to be output from the speaker.

(486) Furthermore, the recorder controller **1226** decodes the obtained encoded data of the EPG data, and supplies the decoded EPG data to the EPG data memory **1227**.

(487) The hard disk recorder **1200** described above includes the image decoding apparatus **500** serving as the video decoder **1225**, the decoder **1252**, and the decoder included in the recorder controller **1226**. That is, the video decoder **1225**, the decoder **1252**, and the decoder included in the recorder controller **1226** appropriately decodes encoded data for which the selection of the I_PCM mode is controlled in units of CUs, which are smaller than LCUs, as in the case of the image decoding apparatus **500**. Accordingly, the video decoder **1225**, the decoder **1252**, and the decoder included in the recorder controller **1226** is capable of realizing a decrease in redundant processing for encoding and a decrease in redundant information included in the encoded data. Accordingly, the video decoder **1225**, the decoder **1252**, and the decoder included in the recorder controller **1226** are capable of realizing enhanced encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(488) Therefore, the hard disk recorder **1200** is capable of realizing enhanced encoding efficiency of data while suppressing a decrease in the efficiency of encoding processing when generating video data (encoded data) received by the tuner or the communication unit **1235** or video data (encoded data) reproduced by the recording/reproducing unit **1233**.

(489) Also, the hard disk recorder **1200** includes the image encoding apparatus **300** serving as the encoder **1251**. Thus, the encoder **1251** controls the selection of the I_PCM mode in units of CUs, which are smaller than LCUs, as in the case of the image encoding apparatus **300**. That is, the encoder **1251** is capable of further reducing redundant processing for encoding and is also capable of reducing redundant information included in encoded data. Accordingly, the encoder **1251** is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(490) Therefore, the hard disk recorder **1200** is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing, for example, when generating encoded data to be recorded on a hard disk.

(491) Note that, although a description has been given above of the hard disk recorder **1200** that records video data and audio data on a hard disk, any types of recording media may of course be used. For example, the image encoding apparatus **300** and the image decoding apparatus **500** can be applied to a recorder that uses a recording medium other than a hard disk, for example, a flash

memory, an optical disc, or video tape, as in the case of the above-described hard disk recorder **1200**.

7. Seventh Embodiment

(492) [Camera]

(493) FIG. **25** is a block diagram illustrating a main example configuration of a camera that includes the image encoding apparatus **300** and the image decoding apparatus **500**.

(494) The camera **1300** illustrated in FIG. **25** captures an image of a subject, causes an LCD **1316** to display the image of the subject, and records it as image data on a recording medium **1333**.

(495) A lens block **1311** causes light (that is, an image of a subject) to enter a CCD/CMOS **1312**. The CCD/CMOS **1312** is an image sensor including a CCD or CMOS, converts the intensity of received light into an electric signal, and supplies the electric signal to a camera signal processor **1313**.

(496) The camera signal processor **1313** converts the electric signal supplied from the CCD/CMOS **1312** into color-difference signals of Y, Cr, and Cb, and supplies them to an image signal processor **1314**. The image signal processor **1314** performs certain image processing on an image signal supplied from the camera signal processor **1313** and encodes, with an encoder **1341**, the image signal under control performed by a controller **1321**. The image signal processor **1314** supplies encoded data which is generated by encoding the image signal to a decoder **1315**. Furthermore, the image signal processor **1314** obtains data to be displayed generated by an on screen display (OSD) **1320**, and supplies the data to the decoder **1315**.

(497) In the foregoing process, the camera signal processor **1313** uses a DRAM (Dynamic Random Access Memory) **1318** connected via a bus **1317** as appropriate, and causes the DRAM **1318** to hold image data, encoded data obtained by encoding the image data, or the like as necessary.

(498) The decoder **1315** decodes encoded data supplied from the image signal processor **1314**, and supplies the image data obtained thereby (decoded image data) to the LCD **1316**. Also, the decoder **1315** supplies the data to be displayed supplied from the image signal processor **1314** to the LCD **1316**. The LCD **1316** combines the image of the decoded image data supplied from the decoder **1315** and the image of the data to be displayed as appropriate, and displays the composite image.

(499) The on screen display **1320** outputs data to be displayed, such as a menu screen made up of symbols, characters, or figures, and icons, to the image signal processor **1314** via the bus **1317** under control performed by the controller **1321**.

(500) The controller **1321** executes various processing operations on the basis of a signal representing the detail of an instruction provided from a user using an operation unit **1322**, and controls the image signal processor **1314**, the DRAM **1318**, an external interface **1319**, the on screen display **1320**, a medium drive **1323**, and so forth via the bus **1317**. Programs, data, and so forth which are necessary for the controller **1321** to execute various processing operations are stored in a flash ROM **1324**.

(501) For example, the controller **1321** is capable of encoding image data stored in the DRAM **1318** and decoding encoded data stored in the DRAM **1318** on behalf of the image signal processor **1314** or the decoder **1315**. At this time, the controller **1321** may perform encoding/decoding processing using a format similar to an encoding/decoding format of the image signal processor **1314** or the decoder **1315**, or may perform encoding/decoding processing using a format incompatible with the image signal processor **1314** or the decoder **1315**.

(502) Also, for example, in a case where an instruction to start printing an image is provided from the operation unit **1322**, the controller **1321** reads out image data from the DRAM **1318** and supplies it to a printer **1334** connected to the external interface **1319** via the bus **1317** to print it.

(503) Furthermore, for example, in a case where an instruction to record an image is provided from the operation unit **1322**, the controller **1321** reads out encoded data from the DRAM **1318** and supplies it to the recording medium **1333** loaded in the medium drive **1323** via the bus **1317** to store it.

(504) The recording medium **1333** is, for example, an arbitrary readable and writable removable medium, such as a magnetic disk, a magneto-optical disc, an optical disc, or a semiconductor memory. Of course, the recording medium **1333** may be a removable medium of any type, and may be a tape device, disc, or memory card. Of course, the recording medium **1333** may be a noncontact IC card or the like.

(505) Also, the medium drive **1323** and the recording medium **1333** may be integrated together, and may be constituted by, for example, a non-transportable storage medium, such as a built-in hard disk drive or an SSD (Solid State Drive).

(506) The external interface **1319** is constituted by, for example, a USB input/output terminal or the like, and is connected to the printer **1334** in the case of printing an image. Also, a drive **1331** is connected to the external interface **1319** as necessary, a removable medium **1332**, such as a magnetic disk, an optical disc, or a magneto-optical disc, is loaded thereto as appropriate, and a computer program read out therefrom is installed into the flash ROM **1324** as necessary.

(507) Furthermore, the external interface **1319** includes a network interface connected to a certain network, such as a LAN or the Internet. The controller **1321** is capable of, for example, reading out encoded data from the DRAM **1318** and supplying it from the external interface **1319** to another apparatus connected via a network, in accordance with an instruction provided from the operation unit **1322**. Also, the controller **1321** is capable of obtaining, via the external interface **1319**, encoded data or image data supplied from another apparatus via a network, and causing the DRAM **1318** to hold it or supplying it to the image signal processor **1314**.

(508) The camera **1300** described above includes the image decoding apparatus **500** serving as the decoder **1315**. That is, the decoder **1315** appropriately decodes encoded data for which the selection of the I_PCM mode is controller in units of CUs, which are smaller than LCUs, as in the case of the image decoding apparatus **500**. Accordingly, the decoder **1315** is capable of realizing a decrease in redundant processing for encoding and a decrease in redundant information included in encoded data. Accordingly, the decoder **1315** is capable of realizing enhanced encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(509) Therefore, the camera **1300** is capable of, for example, realizing enhanced encoding efficiency of data while suppressing a decrease in the efficiency of encoding processing when generating image data generated by the CCD/CMOS **1312**, encoded data of video data read out from the DRAM **1318** or the recording medium **1333**, or encoded data of video data obtained via a network.

(510) Also, the camera **1300** includes the image encoding apparatus **300** serving as the encoder **1341**. The encoder **1341** controls the selection of the I_PCM mode in units of CUs, which are smaller than LCUs, as in the case of the image encoding apparatus **300**. That is, the encoder **1341** is capable of further reducing redundant processing for encoding and is also capable of further reducing redundant information included in encoded data. Accordingly, the encoder **1341** is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing.

(511) Therefore, the camera **1300** is capable of enhancing encoding efficiency while suppressing a decrease in the efficiency of encoding processing, for example, when generating encoded data to be recorded on the DRAM **1318** or the recording medium **1333** or encoded data to be provided to another apparatus.

(512) In addition, the decoding method of the image decoding apparatus **500** may be applied to decoding processing performed by the controller **1321**. Likewise, the encoding method of the image encoding apparatus **300** may be applied to encoding processing performed by the controller **1321**.

(513) Also, the image data captured by the camera **1300** may be a moving image or a still image.

(514) Of course, the image encoding apparatus **300** and the image decoding apparatus **500** can be applied to an apparatus or a system other than the above-described apparatuses.

(515) The present technology can be applied to an image encoding apparatus and an image decoding apparatus which are used to, for example, receive image information (bit stream) compressed through an orthogonal transform such as the discrete cosine transform and by motion compensation via a network medium such as satellite broadcasting, cable TV, the Internet, or a mobile phone, or to process the image information on a storage medium such as an optical or magnetic disk or a flash memory, as in MPEG, H.26x, or the like.

(516) In addition, the present technology may also provide the following configurations.

(517) (1) An image processing apparatus including: an encoding mode setter that sets, in units of coding units having a hierarchical structure, whether a non-compression mode is to be selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and an encoder that encodes the image data in units of the coding units in accordance with a mode set by the encoding mode setter.

(518) (2) The image processing apparatus according to (1), further including: a shift processing controller that performs control, on a coding unit for which the non-compression mode has been set by the encoding mode setter, to skip shift processing in which a bit precision for encoding or decoding is increased; and a shift processor that performs the shift processing on a coding unit of the image data, the coding unit being controlled by the shift processing controller so as to undergo the shift processing.

(519) (3) The image processing apparatus according to (1) or (2), further including: a filter processing controller that performs control, on a coding unit for which the non-compression mode has been set by the encoding mode setter, to skip filter processing in which filtering is performed on a locally decoded image; a filter coefficient calculator that calculates a filter coefficient for the filter processing by using image data corresponding to a coding unit which is controlled by the filter processing controller so as to undergo the filter processing; and a filter processor that performs the filter processing in units of blocks, which are units of the filter processing, by using the filter coefficient calculated by the filter coefficient calculator.

(520) (4) The image processing apparatus according to (3), wherein the filter processor performs the filter processing on only pixels which are controlled by the filter processing controller so as to undergo the filter processing, the pixels being included in a current block which is a target to be processed.

(521) (5) The image processing apparatus according to (3) or (4), further including: a filter identification information generator that generates filter identification information in units of the blocks, the filter identification information being flag information indicating whether the filter processing is to be performed.

(522) (6) The image processing apparatus according to any of (3) to (5), wherein the filter processor performs adaptive loop filtering on the locally decoded image, the adaptive loop filtering being adaptive filter processing using classification processing.

(523) (7) The image processing apparatus according to any of (1) to (6), wherein, in a case where an amount of code of encoded data, which is obtained by encoding the image data corresponding to a current coding unit as a target of encoding processing, is smaller than or equal to an amount of input data, which is a data amount of the image data corresponding to the current coding unit, the encoding mode setter sets an encoding mode of the current coding unit to be the non-compression mode.

(524) (8) The image processing apparatus according to (7), further including: an input data amount calculator that calculates the amount of input data, wherein the encoding mode setter compares, regarding the current coding unit, the amount of input data calculated by the input data amount calculator with the amount of code.

(525) (9) The image processing apparatus according to any of (1) to (8), further including: an identification information generator that generates identification information in units of the coding units, the identification information indicating whether the non-compression mode has been set by

the encoding mode setter.

(526) (10) An image processing method for an image processing apparatus, including: setting, with an encoding mode setter, in units of coding units having a hierarchical structure, whether a non-compression mode is to be selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and encoding, with an encoder, the image data in units of the coding units in accordance with a set mode.

(527) (11) An image processing apparatus including: an encoding mode determiner that determines, in units of coding units having a hierarchical structure, whether a non-compression mode has been selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and a decoder that decodes the encoded data in units of the coding units in accordance with a mode determined by the encoding mode determiner.

(528) (12) The image processing apparatus according to (11), further including: a shift processing controller that performs control, on a coding unit for which the encoding mode determiner has determined that the non-compression mode has been selected, to skip shift processing in which a bit precision for encoding or decoding is increased; and a shift processor that performs the shift processing on a coding unit of the image data, the coding unit being controlled by the shift processing controller so as to undergo the shift processing.

(529) (13) The image processing apparatus according to (11) or (12), further including: a filter processing controller that performs control, on a coding unit for which the encoding mode determiner has determined that the non-compression mode has been selected, to skip filter processing in which filtering is performed on a locally decoded image; and a filter processor that performs the filter processing on the image data in units of blocks, which are units of the filter processing, wherein the filter processor performs the filter processing on only pixels which have been controlled by the filter processing controller so as to undergo the filter processing, the pixels being included in a current block which is a target to be processed.

(530) (14) The image processing apparatus according to (13), wherein the filter processor performs adaptive loop filtering on the locally decoded image, the adaptive loop filtering being adaptive filter processing using classification processing.

(531) (15) The image processing apparatus according to (13) or (14), wherein the filter processor performs the filter processing, in a case where filter identification information indicating whether or not the filter processing has been performed indicates that the filter processing has been performed on image data corresponding to the current block which is a target to be processed, only when control is performed by the filter processing controller so as to perform the filter processing on all pixels included in the current block.

(532) (16) The image processing apparatus according to any of (11) to (15), wherein the encoding mode determiner determines whether the non-compression mode has been selected, on the basis of identification information indicating whether the non-compression mode has been selected in units of the coding units.

(533) (17) An image processing method for an image processing apparatus, including: determining, with an encoding mode determiner, in units of coding units having a hierarchical structure, whether a non-compression mode has been selected as an encoding mode for encoding image data, the non-compression mode being an encoding mode in which the image data is output as encoded data; and decoding, with a decoder, the encoded data in units of the coding units in accordance with a determined mode.

REFERENCE SIGNS LIST

(534) **300** image encoding apparatus **303** adaptive shift-to-left unit **307** lossless encoder **312** loop filter **313** adaptive shift-to-right unit **315** adaptive shift-to-left unit **320** rate controller **321** PCM encoder **331** NAL encoder **332** CU encoder **341** I_PCM_flag generator **342** PCM deciding unit **351**

deblocking filter 352 pixel sorting unit 353 filter coefficient calculator 354 filtering unit 361 input data amount calculator 362 PCM determining unit 363 encoding controller 364 adaptive shift controller 365 filter controller 500 image decoding apparatus 502 lossless decoder 506 loop filter 507 adaptive shift-to-right unit 511 adaptive shift-to-left unit 516 PCM decoder 531 NAL decoder 532 CU decoder 541 I_PCM_flag buffer 542 PCM controller 551 deblocking filter 552 pixel sorting unit 553 filtering unit

Claims

1. An image processing apparatus comprising: a filter configured to perform filter processing in which filtering is performed on a coding unit of a decoded image after a deblocking filter in an encoding process; and a filter processing controller configured to perform control, according to filter identification information indicating whether, on a coding unit for which a non-compression mode has been set, to skip the filter processing in which the filtering is performed on the coding unit of the decoded image after the deblocking filter, the coding unit being formed by recursively splitting a largest coding unit into smaller coding units as block partitioning and the non-compression mode being an encoding mode in which image data is output as encoded data, wherein the filter and the filter processing controller are each implemented via at least one processor.
2. The image processing apparatus according to claim 1, wherein the filter is further configured to perform adaptive loop filtering on the coding unit of the decoded image, the adaptive loop filtering comprising adaptive filter processing using classification processing.
3. An image processing method comprising: performing filter processing in which filtering is performed on a coding unit of a decoded image after a deblocking filter in an encoding process; and controlling, according to filter identification information indicating whether, on the coding unit for which a non-compression mode has been set, to skip the filter processing in which the filtering is performed on the coding unit of the decoded image after the deblocking filter, the coding unit being formed by recursively splitting a largest coding unit into smaller coding units as block partitioning and the non-compression mode being an encoding mode in which image data is output as encoded data.
4. An image processing apparatus comprising: a filter configured to perform filter processing in which filtering is performed on a coding unit of a decoded image after a deblocking filter in a decoding process; and a filter processing controller configured to perform control, according to filter identification information indicating whether, on a coding unit for which a non-compression mode has been set, to skip the filter processing in which the filtering is performed on the coding unit of the decoded image after the deblocking filter, the coding unit being formed by recursively splitting a largest coding unit into smaller coding units as block partitioning and the non-compression mode being an encoding mode in which image data is output as encoded data, to skip the filter processing in which the filtering is performed on the decoded image after the deblocking filter, wherein the filter and the filter processing controller are each implemented via at least one processor.
5. An image processing method comprising: performing filter processing in which filtering is performed on a coding unit of a decoded image after a deblocking filter in a decoding process; and controlling, according to filter identification information indicating whether, on a coding unit for which a non-compression mode has been set, to skip the filter processing in which the filtering is performed on the coding unit of the decoded image after the deblocking filter, the coding unit being formed by recursively splitting a largest coding unit into smaller coding units as block partitioning and the non-compression mode being an encoding mode in which image data is output as encoded data, to skip the filter processing in which the filtering is performed on the decoded image after the deblocking filter.

